Deep Orthogonal Decomposition for surrogate modelling of time dependent PDEs

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Todo list

0.1 Notation

We set $\mathbb{N} := \{1, 2, \dots\}$, $\mathbb{N}_0 := \{0, 1, \dots\}$ and $\mathbb{N}_{\geq n} := \{n, n+1, \dots\}$ for $n \in \mathbb{N}$. Next the set $\mathbb{R}^+ := \{x \in \mathbb{R} \,|\, x \geq 0\}$ will define all real numbers, which are non-negative. Furthermore for any set A, we will write $|A|$ to denote its cardinality. If $x \in \mathbb{R}$, the gaussian ceiling function is given via $\lceil x \rceil := \min\{k \in \mathbb{Z} : k \geq x\}$.

Further if $d \in \mathbb{N}$ and $\|\cdot\|$ is a norm, then for $x \in \mathbb{R}^d$ and $r > 0$ we set $B_{r,\|\cdot\|}(x)$ as the open ball around x with radius r. Additionally $|\cdot|$ denotes the regular euclidean norm on $\mathbb{R}^d$. We usually do not differ between vectors and scalars in ubiquitous notation.

The transpose of a matrix $A \in \mathbb{R}^{d_1 \times d_2}$ for some $d_1, d_2 \in \mathbb{N}$ is given by $A^T \in \mathbb{R}^{d_2 \times d_1}$. The identity matrix will be denoted with $\mathbb{I}_d := \mathrm{diag}(1, \dots, 1) \in \mathbb{R}^{d \times d}$ for any $d \in \mathbb{N}$. Sometimes we abbreviate $\mathbb{I} = \mathbb{I}_d$, when the dimensionality is clear.

Let $(a_n)_{n \in \mathbb{N}} \in \mathbb{R}$ be some sequence, then we say $(a_n)_n$ has *(convergence of) order* n, if there is $c \in \mathbb{R}$ independent of n, s.t. $a_n = c \cdot n$ for all $n \in \mathbb{N}$. We then denote $a_n = \mathcal{O}(n)$.

When writing $a \ll b$ for $a, b \in \mathbb{R}$, the reader can infer, that $a < b$, but also, that the different is "substancial". We do not provide a rigorous definition of this, even though some literature does on occasion.

Moreover, we define the *multiindex* notation as follows: For $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, we write $|\alpha| := \alpha_1 + \cdots + \alpha_d$ and $\alpha! := \alpha_1! \cdots \alpha_d!$. If $x \in \mathbb{R}^d$, then we set

$$x^\alpha := \prod_{i=1}^{d} x_i^{\alpha_i}.$$

0.1.1 Partial Differential Equations

For a function $f : X \to Y$ we denote $\mathrm{supp}(f) := \{x \in X \,|\, f(x) \neq 0\}$ as the *support* of f. Let $g : Y \to Z$ be another function, then $g \circ f : X \to Z$ is their *composition.* Setting the standard topology of $\mathbb{R}^d$ lets us define $\overline{A}$ for some $A \subset \mathbb{R}^d$ as the *closure* of A, and ∂A its *boundary.* If A is non-empty, we set $\mathrm{diam}(A) := \sup_{x,y \in A} |x - y|$ as the diameter of A.

Let $\Omega \subset \mathbb{R}^d$ be open. For $n \in \mathbb{N}_0 \cup \{\infty\}$, we denote by $C^n(\Omega)$ the set of n-times continuously differentiable functions on Ω.

For $f \in C^n(\Omega)$ and $\alpha \in \mathbb{N}_0^d$ with $|\alpha| \leq n$ we write

$$D^\alpha f := \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}.$$

A function $f : \Omega \to \mathbb{R}^m$ for $\Omega \subset \mathbb{R}^d$ is called *Lipschitz continuous*, if there is a constant $L \geq 0$ such that

$$|\, f(x) - f(y) \,| \leq L|x - y|.$$

We further denote with

$$L^p(\Omega) := \left\{ f : \Omega \to \mathbb{R} \,|\, f \text{ is measurable, and } \int_\Omega |f(x)|^p d\mu(x) < \infty \right\}$$

the Lebesgue space and L^p_{loc} as the space of $L^p(\Omega)$ functions with local support. If not specified otherwise we write $\|\cdot\|_p := \|\cdot\|_{L^p(\Omega)}$ for all $1 \leq p \leq \infty$. Note that in critical situations we tend to be explicit about the norms.

0.1.2 Neural Networks

Consider $d_1, d_2 \in \mathbb{N}$ and a matrix $A \in \mathbb{R}^{d_1 \times d_2}$, then the number of non-zero entries of A is given by

$$\|A\|_{\ell^0} := |\{(i,j) \,:\, A_{i,j} \neq 0\}|.$$

Let $B \in \mathbb{R}^{d_1 \times d_3}$ be some other matrix for $d_3 \in \mathbb{N}$. Then we write

$$\begin{bmatrix} A & B \end{bmatrix} \in \mathbb{R}^{d_1 \times (d_2+d_3)} \quad \text{or} \quad \left[\begin{array}{c|c} A & B \end{array}\right] \in \mathbb{R}^{d_1 \times (d_2+d_3)},$$

for the stacking of matrices, where the latter notation is meant to symbolize a strong delineation between blocks. Similar applies to vertical stacking, if $A \in \mathbb{R}^{d_1 \times d_2}$ and $B \in \mathbb{R}^{d_3 \times d_2}$. As a special case, this applies to vectors as well.

Chapter 1

Introduction

Write an Introduction

1.1 Time Dependent PDEs

The study and analysis of partial differential equations (PDEs) began with the works of Euler, d'Alembert, Lagrange and Laplace (see (Brezis & Browder, 1998) for this introduction). As so often in mathematics it started off with a set of simple problems, such as the vibrating string in one dimension as modelled by d'Alembert (1752)

$$\frac{\partial^2}{\partial t^2}u(x,t) = \frac{\partial^2}{\partial x^2}u(x,t), \quad \text{for } u \in C^2(\mathbb{R} \times \mathbb{R}^+), x \in \mathbb{R} \text{ and } t \in \mathbb{R}^+,$$

the extension to three dimensions by Euler (1759) and Bernoulli (1762), the Laplace equation for gravitational fields (1780)

$$\Delta u(x) = 0, \quad \text{for } u \in C^2(\mathbb{R}^3) \text{ and } x \in \mathbb{R}^3,$$

and finally the heat equation, as introduced by Fourier (1810 - 1822)

$$\frac{\partial}{\partial t}u(x,t) = \Delta u(x,t), \quad \text{for } u \in C^2(\mathbb{R}^3 \times \mathbb{R}^+), x \in \mathbb{R}^3 \text{ and } t \in \mathbb{R}^+.$$

As any posed problem calls for a solution to it, there have been simultaneous developments with regard to the calculation methods; the method of seperation of variables was given by Euler and d'Alembert for the study of spherical harmonics and Fourier introduced similar ideas for the heat equation. For the Laplace equation, the Green functions were introduced in 1835. Green also discovered an important principle for solvability of the (later so called) Dirichlet problem

$$\begin{cases} \Delta u(x) = 0, & \text{for } x \in \Omega \subset \mathbb{R}^3, \\ u(x) = g(x), & \text{for } x \in \partial\Omega, \end{cases}$$

for some function $g \in C^0(\mathbb{R}^3)$, which revolved around minimizing an energy functional $E : C^2(\mathbb{R}^3) \to \mathbb{R}^+$, thus founding the variational principle.

Though these were first attempts at solving such problems, the idea of Green in particular led to a few analytical problems, by the lack of mathematical rigor. Together with the contribution of Cauchy on the *initial value* problem, this laid the groundwork for the field of *Existence Theory*, which tries to establish existence results for PDEs.

Notable here are the alternating method of Schwarz (1870) using the idea of domain decomposition and especially Neumanns method of integral equations developed in 1877, which would go on to be the starting point for the weak form and subsequently the Sobolev spaces. These were a necessity after B. Levi showed in 1906, that a general minimizing sequence for the Dirichlet integral was a Cauchy sequence and would therefore converge in the right space, which was then given due to the Sobolev embedding theorems.

To tackle the problem arising from "exotic" examples and the complicated nature of some PDEs, Hadamard proposed a classification of the most common linear PDEs using its characteristic polynomial, which is given by replacing all partial derivatives $\partial/\partial x_j$ by the algebraic variable ξ_j. In that case the PDE is called

1. *elliptic*, if after a change of varibles the polynomial can be written as $\xi_1^2 + \xi_2^2 + \cdots + \xi_n^2$,
2. *parabolic*, if after a change of varibles the polynomial can be written as $\xi_1 + \xi_2^2 + \cdots + \xi_n^2$,
3. *hyperbolic*, if after a change of varibles the polynomial can be written as $\xi_1^2 - (\xi_2^2 + \cdots + \xi_n^2)$.

These analytical results can be expanded a lot more, but what is most interesting in terms of this paper, are the Sobolev spaces. They, and the closely connected Bochner spaces, are the relevant solution spaces to the limits of all, in some sense, consistent numerical schemes, since the solutions are at least embedded in them. This follows quite natural, as any numerical scheme is basically bounded by the necessity of being ran on a computer and thus having a finite dimensional space it defines its approximation on, and at the very least the Sobolev and Bochner spaces admit any piecewise smooth function.

1.2 Neural networks

1.3 Motivation

1.4 Literature review

1.5 Outline

Chapter 2

Theoretical Background

2.1 Linear Algebra

As preparation for the comming Bauer Fike Theorem, we want to define a matrix norm for the space $\mathbb{R}^{N\times N}$, where $N \in \mathbb{N}$, as follows

$$\|A\|_{\mathrm{op}} := N \cdot \max_{1\leq i,j\leq N} |\,[A]_{ij}\,|\,.$$

This satifies the wanted property mentioned in the origional paper (Bauer & Fike, 1960, p. 137-141); any operator norm such that

$$\|Ax\| \leq \|A\|_{\mathrm{op}}\|x\|, \quad \text{for all } x \in \mathbb{R}^N$$

for some norm $\|\cdot\|$ on $\mathbb{R}^N$ can be used for the statement of the following theorem:

Theorem 2.1 (Bauer-Fike Theorem)**.** *Let $K \in \mathbb{R}^{N\times N}$ be some diagonalizable matrices with σ_k eigenvalues and $V = (\xi_1, \dots, \xi_N) \in \mathbb{R}^{N\times N}$ be the stacked eigenvectors matrix of K and $\delta K \in \mathbb{R}^{N\times N}$ some symmetric matrix. Then if λ_k are the eigenvalues of $K + \delta K$, we have*

$$\min_k |\lambda_k - \sigma_k| \leq \kappa(V)\|(K + \delta K) - K\|_{op},$$

where $\kappa(V) = \|V\|_{op}\|V^{-1}\|_{op}$.

add prove

2.2 Parametric Partial Differential Equations

2.2.1 Sobolev Spaces

For our work with Partial Differential Equation, and subsequent error approximations, we want to introduce Sobolev spaces. These are generalized differentiable spaces, where a weak derivative is given, as an analogon to the classic derivative in the $C^n(\Omega)$ space. They are employed in order to exploit their compact embedding in the Hölder spaces, which are generalizations of $C^n(\Omega)$. We will not go in much detail about the theory of Sobolev spaces, since we mostly concern ourselves with finite dimensional subspaces, but some ground truth must be discussed.

For this subsection we primary use (Alt, 2016), and for the first upcoming definition (Dacorogna, 2004).

Definition 2.2 (Lipschitz Domain). Let $d \in \mathbb{N}$, and $\Omega \subset \mathbb{R}^d$. Then Ω is called a *Lipschitz domain* if for every $x \in \partial\Omega$ and some $r > 0$, there is $U := B_{r,|\cdot|}(x) \subset \mathbb{R}^d$ and an isomorphic map $H : Q \to B_{r,|\cdot|}(x)$ with $Q := \{x \in \mathbb{R}^d \mid \|x\|_{\ell^1} < d\}$ such that

$$H \in C^k(\overline{Q}), \quad H^{-1} \in C^k(\overline{U}), \quad H(Q_+) = U \cap \Omega, \quad H(Q_0) = U \cap \partial\Omega$$

where $Q_+ := \{x \in Q \mid x_d > 0\}$ and similarily $Q_0 := \{x \in Q \mid x_d = 0\}$, and H is only in $C^{0,1}$.

From now on, let $d \in \mathbb{N}$ and, if not given otherwise, $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain.

Definition 2.3 (Weak Derivative). Let $\alpha \in \mathbb{N}_0^d$ be a multiindex. A function $u \in L^1_{\text{loc}}(\Omega)$ is said to have an *α-th weak derivative*, if there is a function $v_\alpha \in L^1_{\text{loc}}(\Omega)$ such that

$$\int_\Omega u D^\alpha \varphi = (-1)^{|\alpha|} \int_\Omega v_\alpha \varphi \quad \forall \varphi \in C_0^\infty(\Omega).$$

We then write $v_\alpha = D^\alpha u$.

Definition 2.4 (Sobolev Space). Let $n \in \mathbb{N}_0$ and $1 \leq p \leq \infty$. Then we define the *Sobolev space* as

$$W^{n,p}(\Omega) := \{f \in L^p(\Omega) \mid \|f\|_{W^{n,p}(\Omega)} < \infty\},$$

where the norm on this space is given via

$$\|f\|_{W^{n,p}(\Omega)} := \begin{cases} \left(\sum_{|\alpha| \leq n} \|D^\alpha f\|^p_{L^p(\Omega)} \right)^{1/p}, & \text{if } p < \infty \\ \max_{|\alpha| \leq n} \|D^\alpha f\|_{L^\infty(\Omega)}, & \text{if } p = \infty, \end{cases}$$

for any $f \in W^{n,p}(\Omega)$.

It is possible to extend this definition for $0 < n < 1$ to that of the *Sobolev-Slobodeckij spaces*. In the analysis of partial differential equations these are used to characterize the regularity of Sobolev functions, which are restricted to the boundary $\partial\Omega$. This restriction itself is called a trace operator. One of the statements of relevance for such fractal Sobolev spaces is the following:

If $\Omega \subset \mathbb{R}^d$ is a regular open set (for example, if Ω is a Lipschitz domain), and if $u \in W^{1,p}(\Omega)$, then in fact $u \mapsto u|_{\partial\Omega}$ as a trace operator is surjective from $W^{1,p}(\Omega)$ onto $W^{1-(1/p),p}(\partial\Omega)$ and

$$\|u|_{\partial\Omega}\|_{W^{1-(1/p),p}(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)},$$

for some $C > 0$ independent of p and u, see (Brezis, 2011, p. 315). This of course only makes sense in light of the next definition.

Definition 2.5 (Sobolev-Slobodeckij Space). Let $0 < s < 1$ and $1 \leq p \leq \infty$. Then we define the *Sobolev-Slobodeckij space* as

$$W^{s,p}(\Omega) := \{f \in L^p(\Omega) \mid \|f\|_{W^{s,p}(\Omega)} < \infty\},$$

where the norm on this space is given via

$$\|f\|_{W^{s,p}(\Omega)} := \begin{cases} \left(\|f\|^p_{L^p(\Omega)} + \int_\Omega \int_\Omega \left(\frac{|f(x) - f(y)|}{|x - y|^{s+d/p}} \right)^p dx dy \right)^{1/p}, & \text{if } p < \infty \\ \max \left\{ \|f\|_{L^\infty(\Omega)}, \operatorname*{ess\,sup}_{x,y \in \Omega} \frac{|f(x) - f(y)|}{|x - y|^s} \right\}, & \text{if } p = \infty, \end{cases}$$

for any $f \in W^{s,p}(\Omega)$.

Theorem 2.6 (Sobolev Embedding Theorem)**.** *Let $m_1, m_2 \in \mathbb{N}_0$, $1 \leq p < \infty$ and $\alpha \in (0,1)$. If we choose*

$$m_1 - \frac{d}{p} = m_2 + \alpha,$$

then the identity defines a continuous embedding

$$W^{m_1,p}(\Omega) \hookrightarrow C^{m_2,\alpha}(\overline{\Omega}).$$

If even $m_1 - \frac{d}{p} > m_2 + \alpha$, then the embedding is compact.

Proof. See (Alt, 2016, p. 10-13). □

2.2.2 Existence and Uniqueness

This subsection provides a brief introduction to one specific statement of existence and uniqueness, which we coincidentally can apply well to the projected version of our finite dimensional problem. Note, that of course, this does not justify any existence result for highly non-linear equations on a lot of upstream classical or even weak problems, like the famous incompressible Navier-Stokes Equation.

We start by defining the differential operator class, to which we can then apply the Hille-Yosida Theorem. Most notable, we know with that subclass, that any monotone operator, which acts on a finite dimensional space, is in that class, as will be shown afterwards.

Definition 2.7 (Maximal Monotone Operator)**.** Let $A : D(A) \subset H \to H$ be an unbounded linear operator. It is *monotone*, if

$$\langle Av, v\rangle_H \geq 0, \quad \text{for all } v \in D(A),$$

and *maximal monotone*, if additionally for all $f \in H$ there is $u \in D(A)$ such that

$$u + Au = f.$$

Lemma 2.8. Let H be a finite dimensional real Hilbert space, and $A : H \to H$ a monotone operator according to Definition (2.7). Then A is maximal monotone.

Proof. By obvious identification with a basis, we know, that $H \cong \mathbb{R}^N$ for some $N \in \mathbb{N}$. Thus any element $u \in H$ can be expressed as

$$u = \sum_{j=1}^{N} u_j \varphi_j,$$

where $u_j \in \mathbb{R}$ and $\varphi_j \in H$ for all $j = 1, \ldots, N$, and similarily for $i, j, k = 1, \ldots, N$ there are $a_{ik} \in \mathbb{R}$ such that

$$Au = \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{k=1}^{N} a_{ik} u_j \langle \varphi_j, \varphi_i\rangle \varphi_k.$$

Thus the condition for A being maximal monotone simplyfies to finding $[u_j]_{j=1}^N \in R^N$ for an arbitrary $[f_j]_{j=1}^N \in \mathbb{R}^N$ where

doesnt work

$$u_j + \sum_{i=1}^{N} a_{ij} u_j = f_j$$

for all $j = 1, \dots, N$, which can obviously be done in general. □

Theorem 2.9 (Hille-Yosida Theorem). *Let A be a maximal monotone operator, H some Hilbert space and $D(A) \subset H$. Then, given any $u_0 \in D(A)$ there is a unique function*

$$u \in C^1([0,\infty); H) \cap C([0,\infty); D(A))$$

satisfying

$$\begin{cases} \dfrac{\partial u}{\partial t} + Au = 0 & \text{on } [0,\infty), \\ u(0) = u_0 & \text{on } \{0\}. \end{cases}$$

Moreover,

$$\|u(t)\|_H \leq \|u_0\|_H,$$

for all $t \geq 0$.

Proof. See (Brezis, 2011, p. 185-191). □

2.2.3 Reduced Basis Framework

Give a reasonable weak formulation framework to appreciate gram matrix

Introduce the general notion of the POD

Introduce KnW and its relation to the continous case

For this subsection we define $u(\mu_j, t_k) \in \mathbb{R}^{N_h}$ for $(\mu_j)_{j=1}^{N_s} =: \mathcal{P} \subset \Theta$ and $(t_k)_{k=1}^{N_t} =: \mathcal{T} \subset \mathbb{T}$ to be some vectors, later representing the solutions to a parametric PDE for a given parameter and time point in a given discrete subspace of the coefficient solution manifold. Here Θ represents and arbitrary parameter space, $\mathbb{T}$ some arbitrary time space, and $N_s, N_t \in \mathbb{N}$ be the parameter and time sample size respectively. Let the norm on $\mathbb{R}^{N_h}$ be defined here, using a symmetric positive definite matrix $\mathbb{G} \in \mathbb{R}^{N_h \times N_h}$, as follows

$$\|u(\mu, t)\| := \sqrt{u(\mu, t)^T \, \mathbb{G} \, u(\mu, t)}.$$

This is meant to result from a choice of an inner product stemming from the coefficient vectors $u(\mu, t)$ representing that space, in other words, given by the choice of a basis on the corresponding Hilbert space, when viewing $u(\mu, t) \in \mathbb{R}^{N_h}$ as the coefficients identifying an element in that space. It is assumed, that we sample the parameters and time points uniformly and iid from their respective spaces and that $\|u(\mu, t)\| < \infty$ for all $\mu \in \Theta$ and $t \in \mathbb{T}$. Let also

$$N_{\text{data}} := N_s \cdot N_t,$$

be the total sample size.

Here we will discuss prerequisites to the Proper Orthogonal Decomposition and related concepts, such as the Kolmogorov n-width, which lets us derive many key concepts and is an important quantity in the field of ROM methods. We base this subsection on (Quarteroni et al., 2015), but import the generalizations to arbitrary inner products from (Franco et al., 2024). Note, that in practice, the performing of a POD for each μ seperately would be unfeasible, but we want to keep the choice of basis parameter sensitive, as it will open the door for some interesting facts down the road.

Definition 2.10 (Generalized Proper Orthogonal Decomposition)**.** Assuming $N_{\text{data}} < \infty$ and let $\mathbb{G} \in \mathbb{R}^{N_h \times N_h}$ be a mass matrix, we define by $U_j = [u(\mu_j, t_k)]_{k=1}^{N_t} \in \mathbb{R}^{N_h \times N_t}$ the snapshot matrix for a given $\mu_j \in \mathcal{P}$ and consequently $U = [U_j]_{j=1}^{N_s} \in \mathbb{R}^{N_h \times N_{\text{data}}}$ the collective snapshot matrix. Then the *(discrete) correlation matrix* is given by

$$K = \frac{|\mathcal{P} \times \mathcal{T}|}{N_{data}} U \mathbb{G} U^T \in \mathbb{R}^{N_h \times N_h}. \tag{2.1}$$

K is symmetric and semi-positive definite and thus has real and positive eigenvalues $\sigma^2 \geq \cdots \geq \sigma_r^2$. We further call ψ_k^μ the eigenvectors of the correlation matrix K. With these, the corresponding POD basis of dimension $N < N_h$ is given via

$$\xi_k = \frac{1}{\sigma_k} U \psi_k, \quad 1 \leq k \leq N,$$

for the N largest eigenvalues and their corresponding eigenvectors. Then stacking these eigenvectors gives us the *POD matrix*

$$\mathbb{A} := [\xi_k]_{k=1}^N \in \mathbb{R}^{N_h \times N}.$$

This matrix is $\mathbb{G}$-orthonormal by construction. Further, if $N_s, N_t = \infty$, we define the *continuous correlation matrix* via

$$K_\infty = \int_{\mathcal{P} \times \mathcal{T}} u(\mu, t) \mathbb{G} u^T(\mu, t) d(\mu, t) \in \mathbb{R}^{N_h \times N_h}, \tag{2.2}$$

and its real eigenvalues by $\sigma_{k,\infty}^2$.

Algorithm 1 Generalized Proper Orthogonal Decomposition (POD)

Input: snapshot matrix $U = [U_j]_{j=1}^{N_s} \in \mathbb{R}^{N_h \times N_{\text{data}}}$, mass matrix $\mathbb{G} \in \mathbb{R}^{N_h \times N_h}$, reduction $N < N_h$
Output: POD basis matrix $\mathbb{A}^\mu = [\xi_1^\mu, \ldots, \xi_N^\mu] \in \mathbb{R}^{N_h \times N}$
if $N_{\text{data}} \leq N_h$ **then**
  Form the correlation matrix $C = \frac{|\mathcal{P} \times \mathcal{T}|}{N_{data}} U^T \mathbb{G} U \in \mathbb{R}^{N_{\text{data}} \times N_{\text{data}}}$
  Solve the eigenvalue problem $C\psi_i = \sigma_i^2 \psi_i, \quad i = 1, \ldots, r$
  Set $\xi_i = \frac{1}{\sigma_i} U \psi_i, \quad i = 1, \ldots, r$
else
  Form the correlation matrix $K = \frac{|\mathcal{P} \times \mathcal{T}|}{N_{data}} U \mathbb{G} U^T \in \mathbb{R}^{N_h \times N_h}$
  Solve the eigenvalue problem $K\xi_i = \sigma_i^2 \xi_i, \quad i = 1, \ldots, r$
end if
$\mathbb{A} \leftarrow \text{stack}(\xi_1, \ldots, \xi_N)$

include theorem 6.1 from quarteroni2015rb

Proposition 2.11. Let K and K_∞ and their respective eigenvalues be defined as in Definition (2.10), then letting $N_s, N_t \to \infty$, we get

$$\sum_{k>N} \sigma_k^2 \longrightarrow \sum_{k>N} \sigma_{k,\infty}^2, \quad \text{a.s.}$$

Proof. Let $N_s, N_t < \infty$, conforming to the sampling assumption, and

$$M := \operatorname*{ess\,sup}_{(\mu,t)\in\Theta\times\mathcal{T}} \|u(\mu,t)\| < \infty$$

as before (see introduction in this Section (2.2.3)). Taking a variational approach on this finite dimensional space, we know, that there exists a $\mathbb{G}$-orthognormal matrix $V_\infty \in \mathbb{R}^{N_h\times N}$ such that

$$\begin{aligned}\sum_{k>N}\sigma_{k,\infty}^2 &= \int_{\mathcal{P}\times\mathcal{T}} \|u(\mu,t) - V_\infty V_\infty^T \mathbb{G}u(\mu,t)\|^2 d(\mu,t)\\ &= \min_{W\in\mathbb{R}^{N_h\times N}: W^T\mathbb{G}W=I} \int_{\mu\times\mathcal{T}} \|u(\mu,t) - WW^T\mathbb{G}u(\mu,t)\|^2 d(\mu,t).\end{aligned}$$

Thus, since this is the optimal choice of an approximation matrix for the continuous case and $\mathbb{A}\in\mathbb{R}^{N_h\times N}$ given by Definition (2.10) is only optimal in the discrete setting and not necessarily the continuous case, we have

$$\sum_{k>N}\sigma_{k,\infty}^2 \leq \int_{\mathcal{P}\times\mathcal{T}} \|u(\mu,t) - \mathbb{A}\mathbb{A}^T\mathbb{G}u(\mu,t)\|^2 d(\mu,t) = \sum_{k>N}\sigma_k^2.$$

We then write entry-wise for $m, l = 1, \dots N_h$

$$\begin{aligned}[K_\infty - K]_{ml} := &\int_{\mathcal{P}\times\mathcal{T}} [\mathbb{G}^{1/2}u(\mu,t)]_m [\mathbb{G}^{1/2}u(\mu,t)]_l d(\mu,t)\\ &- \frac{|\mathcal{P}\times\mathcal{T}|}{N_{data}} \sum_{j=1}^{N_s}\sum_{k=1}^{N_t} [\mathbb{G}^{1/2}u(\mu_j,t_k)]_m [\mathbb{G}^{1/2}u(\mu_j,t_k)]_l.\end{aligned}$$

By the uniform iid sampling assumed and the general bound on $\|u(\mu,t)\| \leq M$ before, we know that with the entry-wise boundedness of the PDE solutions, the resulting matrix is subject to the strong law of large numbers, and thus converges to zero, given the entry-wise matrix norm $\|\cdot\|_{\text{op}}$ as $N_t, N_s \to \infty$ almost surely.

Now using Bauer-Fike's Theorem (2.1) with the $\|\cdot\|_{\text{op}}$-norm, we have upon ordering, that for each $\sigma_{k,\infty}^2$ there is a σ_k^2 in the spectrum of K, such that

$$|\sigma_k^2 - \sigma_{k,\infty}^2| \leq \kappa(\mathbb{A}_\infty)\|K - K_\infty\|_{\text{op}},$$

where $\kappa(\mathbb{A}_\infty)$ denotes the condition number of the stacked eigenvector matrix of K_∞, i.e. $A_\infty = (\xi_{1,\infty}, \dots, \xi_{N_h,\infty})$. This number is independent of N_s and N_t. Now by convergence of $\|K - K_\infty\|_{\text{op}} \to 0$ almost surely for $N_s, N_t \to \infty$, we conclude the proof. □

Proposition 2.12. Let $\mathcal{V}_N = \{W \in \mathbb{R}^{N_h\times N} \,|\, W^T\mathbb{G}W = I_N\}$ be the set of all N-dimensional $\mathbb{G}$-orthonormal bases, $N_{data} := N_tN_s$ and $\mathbb{A} = [\xi]_{i=1}^N$ be the POD matrix according to Definition (2.10), then

$$\begin{aligned}&\frac{|\mathcal{P}\times\mathcal{T}|}{N_{data}} \sum_{i\in I, k=1,\dots,N_t} \|u(\mu_j,t_k) - \mathbb{A}\mathbb{A}^T\mathbb{G}u(\mu_j,t_k)\|_2^2\\ &\qquad = \min_{W\in\mathcal{V}_N} \frac{|\mathcal{P}\times\mathcal{T}|}{N_{data}} \sum_{i\in I,k=1,\dots,N_t} \|u(\mu_j,t_k) - WW^T\mathbb{G}u(\mu_j,t_k)\|_2^2 = \sum_{j=N+1}^{r}\sigma_k^2.\end{aligned}$$

Definition 2.13 (Linear Kolmogorov N-width)**.** Let $N\in\mathbb{N}$, $N \leq N_h$ and $\mathcal{S}_{N_h} = \{u(\mu,t) \in \mathbb{R}^{N_h} \,|\, (\mu,t)\in\mathcal{P}\times\mathcal{T}\}$ be some parametric manifold. The linear Kolmogorov N-width of $\mathcal{S}_{N_h}$ is defined as

$$d_N(S_{N_h}) = \inf_{V\in\mathbb{R}^{N_h\times N}} \sup_{u(\mu,t)\in S_{N_h}} \|u(\mu,t) - VV^T\mathbb{G}u(\mu,t)\|.$$

Definition 2.14 (Nonlinear Kolmogorov N-width)**.** Let $N \in \mathbb{N}$, $N \leq N_h$ and $\mathcal{S}_{N_h} = \{u(\mu,t) \in \mathbb{R}^{N_h} \mid (\mu,t) \in \mathcal{P} \times \mathcal{T}\}$ be some parametric manifold. The nonlinear Kolmogorov N-width of $\mathcal{S}_{N_h}$ is defined as

$$\delta_N(\mathcal{S}_{N_h}) = \inf_{\psi \in C(\mathbb{R}^{N_h},\mathbb{R}^N), \psi' \in C(\mathbb{R}^N,\mathbb{R}^{N_h})} \sup_{u \in \mathcal{S}_{N_h}} |u - \psi(\psi'(u))|.$$

2.3 Neural Networks

In this section, we will introduce the general notion of neural networks. Specifically important for us will be concatenations of perceptrons, which are a specific subclass of networks, the so called *feedforward neural network*.

Definition 2.15 (Neural Network)**.** Let $d, L \in \mathbb{N}$, then a *neural network* Φ *with input dimension* d *and* L *layers* is defined via

$$\Phi = ((A_1, b_1), (A_2, b_2), \dots, (A_L, b_L)),$$

such that $N_0 = d$ and $N_1, \dots, N_L \in \mathbb{N}$, where each A_l is an $N_l \times \sum_{k=0}^{l-1} N_k$ dimensional matrix, and $b_l \in \mathbb{R}^{N_l}$.

We further define $R_\rho(\Phi) : \mathbb{R}^d \to \mathbb{R}^{N_L}$ as a *realizations of* Φ *with activation function* ρ for some arbitrary map $\rho : \mathbb{R} \to \mathbb{R}$ as the result of the following scheme for some input $x \in R^d$:

$$\begin{aligned}
x_0 &:= x, \\
x_l &:= \rho\left(A_l \left[x_0^T \,\middle|\, \cdots \,\middle|\, x_{l-1}^T\right]^T + b_l\right), \quad \text{for } l = 1, \dots, L-1, \\
x_L &:= A_L \left[x_0^T \,\middle|\, \cdots \,\middle|\, x_{l-1}^T\right]^T + b_L,
\end{aligned}$$

where ρ acts componentwise, i.e. $\rho(y) := [\rho(y_1), \dots, \rho(y_m)]^T \in \mathbb{R}^m$ for each $y = [y_1, \dots, y_m] \in \mathbb{R}^m$. Note that we can write

$$A_l = \left[A_{l,x_0} \,\middle|\, \cdots \,\middle|\, A_{l,x_{l-1}}\right],$$

with A_{l,x_k} being a $N_l \times N_k$ dimensional matrix for $k = 0, \dots, l-1$ and $l = 1, \dots, L$. This comes in handy to portray the scheme via

$$\begin{aligned}
x_l &:= \rho\left(A_{l,x_0} x_0 + \cdots + A_{l,x_{l-1}} x_{l-1} + b_l\right), \quad \text{for } l = 1, \dots, L-1, \\
x_L &:= A_{L,x_{L-1}} x_{L-1} + b_L.
\end{aligned}$$

Moreover call $N_\Phi := d + \sum_{l=1}^L N_l$ the *number of neurons* of the network, $L_\Phi := L$ the *number of layers*, and finally $\omega_\Phi := \sum_{l=1}^L (\|A_l\|_{\ell^0} + \|b_l\|_{\ell^0})$ the *number of active weights*.

Remark 2.16. As a note, we can rewrite known neural network architectures, such as a convolutional autoencoder, as a neural network according to Definition (2.15), see also (Gühring et al., 2020, p. 22).

Motivate this with some example.

From this we can infer a simpler architechture, describing a network which omits any connection of non-neighboring layers.

Definition 2.17 (Standard Neural Network)**.** If a neural network $\Phi = ((A_1, b_1), \ldots, (A_L, b_L))$ now has the characteristic

$$A_{l,x_i} = 0, \quad \text{for } l = 1, \ldots, L, \text{ and } i = 0, \ldots, l-2,$$

then we call Φ a *standard neural network*.

When considering the similarities and differences between two distinct neural networks $\Phi_1 = ((A_1, b_1), \ldots, (A_L, b_L))$ and $\Phi_2 = ((\hat{A}_1, \hat{b}_1), \ldots, (\hat{A}_L, \hat{b}_L))$, one can easily see, that if $[A_1]_{ij} = 0$, while $[\hat{A}_1]_{ij} \neq 0$ for some i, j, then these networks are severely different, whereas if they only differ in the specific non-zero weight, they could be recovered from each other through training. This inspires the definition of *an architecture of a network*.

Definition 2.18 (Neural Network Architecture)**.** For some $d, L \in \mathbb{N}$, we define a *neural network architecture $\mathcal{A}$ with input dimension d and L layers* to be a binary neural network, i.e. for all entries of $A_i, b_i \in \{0, 1\}$ for $i = 1, \ldots, L$ of $\mathcal{A}$.

We further say that a *neural network* $\Phi = ((\hat{A}_1, \hat{b}_1), \ldots, (\hat{A}_L, \hat{b}_L))$ has *architecture* $\mathcal{A}$, if

1. $N_l(\Phi) = N_l$ for all $l = 1, \ldots, L$, and
2. $[\hat{A}_l]_{ij} \neq 0$ implies $[A_l]_{ij} \neq 0$ for all $l = 1, \ldots, L$ with $i = 1, \ldots, N_l$ and $j = 1, \ldots, \sum_{k=0}^{l-1} N_k$.

Since the associated realization of a neural network $R_\rho(\Phi)$ requires the definition of ρ, the activation function, we will define the most commonly used one in literature. It will help a lot with some of the constructions, since it has a simple nature, but also provides the network with the necessary non-linearity.

Definition 2.19 (ReLU)**.** The function

$$\rho : \mathbb{R} \to \mathbb{R}; \quad x \mapsto \max(0, x),$$

is called *ReLU* (Rectified Linear Unit) *activation function.*

Given such an activation function, Kurt Hornik provided in 1990 the first contribution to a field, which would emerge centuries later: Approximation theory. He proved for any $L^p(\mu)$ function, for μ some finite measure, the existence of an arbitrarily close ReLU valued neural network (Hornik, 1991).

Theorem 2.20 (Hornik's Theorem)**.** *Define for some continuous, unbounded and non-constant actiation function ρ the set of all feedforward neural networks, as*

$$\mathcal{N}_d(\rho) := \bigcup_{L=1}^{\infty} \left\{ R_\rho(\Phi) : \mathbb{R}^d \to \mathbb{R} \,\middle|\, \Phi \textit{ is neural network with } L \textit{ layers} \right\}.$$

Then $\mathcal{N}_d(\rho)$ lies dense in $L^p(\mu)$ for every finite measure μ, $p \in (0, \infty)$.

The proof of this statement leverages on Hahn Banach, but thus does not provide the specific architecture of the required neural network, which would be used for such an approximation. We aim significantly higher, namely at a bound on the necessary non-zero weights and layers for such an approximation, and will thus not mention the proof beyond this outline.

2.3.1 Network concatenations

In this subsection we will discuss the concatenations of multiple neural networks and the effects on the characteristic numbers of it. The complexity of a concatenated network can be given nicely, when the used activation function is

Definition 2.21 (Network Concatenation)**.** Let $L_1, L_2 \in \mathbb{N}$ and

$$\Phi^1 = ((A_1, b_1), \dots, (A_{L_1}, b_{L_1})), \quad \Phi^2 = ((B_1, c_1), \dots, (B_{L_2}, c_{L_2})),$$

be two neural networks such that the input layer of Φ^1 has the same dimension as the output layer of Φ^2. Then, $\Phi^1 \bullet \Phi^2$ denotes the following $L_1 + L_2 - 1$ layered network:

$$\Phi^1 \bullet \Phi^2 := \Big(((B_1, c_1), \dots, (B_{L_2-1}, c_{L_2-1})), ((\widetilde{A}_1, \widetilde{b}_1), \dots, (\widetilde{A}_{L_1}, \widetilde{b}_{L_1})) \Big),$$

where

$$\widetilde{A}_l^1 := \Big[A_{l,x_0} \, B_{L_2} \; \Big| \; A_{l,x_1} \; \Big| \; \cdots \; \Big| \; A_{l,x_{l-1}} \Big], \quad \text{and} \quad \widetilde{b}_l^1 := A_{l,x_0} \, c_{L_2} + b_l,$$

for $l = 1, \dots, L_1$. We call $\Phi^1 \bullet \Phi^2$ the *concatenation* of Φ^1 and Φ^2.

Lemma 2.22 (Neutral Neural Network)**.** Let $\Phi^{\mathrm{Id}} = ((A_1, b_1), (A_2, b_2))$ be a d-dimensional neural network, with

$$A_1 := \begin{bmatrix} \mathrm{Id}_{\mathbb{R}^d} \\ -\mathrm{Id}_{\mathbb{R}^d} \end{bmatrix}, \quad b_1 := 0, \quad A_2 := \Big[0_{\mathbb{R}^{d \times d}} \; \Big| \; \mathrm{Id}_{\mathbb{R}^d} \; \Big| \; -\mathrm{Id}_{\mathbb{R}^d} \Big], \quad b_2 := 0.$$

Then $R_\rho(\Phi^{\mathrm{Id}}) = \mathrm{Id}_{\mathbb{R}^d}$.

Lemma 2.23. For any two neural networks Φ^1 and Φ^1 according to Definition (2.21), $\Phi^1 \odot \Phi^2 := \Phi^1 \bullet \Phi^{\mathrm{id}} \bullet \Phi^2$ is a neural network with d-dimensional input, $L_1 + L_2$ layers with $\omega_{\Phi^1 \odot \Phi^2} \leq 2\omega_{\Phi^1} + 2\omega_{\Phi^2}$ active weights, $N_{\Phi^1 \odot \Phi^2} \leq 2N_{\Phi^1} + 2N_{\Phi^2}$ neurons. Additionally for all $x \in \mathbb{R}^d$, we have

$$R_\rho(\Phi^1 \odot \Phi^2)(x) = R_\rho(\Phi^1) \circ R_\rho(\Phi^2)(x).$$

Proof.

maybe include proof?

Intuitively speaking the latter network must account for all "missing" links, which a direct concatenation would necessarily be missing; any layer after the "switch" from the first to the latter needs to have a correction term in the weights, additionally attributing all potential connections which the first network would impose.

Let w.l.o.g. $L_2 = 2$, since the pass through the first layers of the combined network, is equivalent to the pass through the first layers of the individual network. Then we can show by induction:

1. Base case: $L_2 = 1$. We argue with $A_{L_1} x = A_{L_1, x_0} \, x$ for x of output dimension of Φ^1, since the input dimension of Φ^1 is the output of Φ^2.

$$\begin{aligned} R_\rho(\Phi^1) \circ R_\rho(\Phi^2)(x_0) &= A_{L_1} \left(B_{L_2, x_0} x_0 + B_{L_2, x_1} x_1 + c_{L_2} \right) + b_{L_1} \\ &= A_{L_1, x_0} B_{L_2, x_0} x_0 + A_{L_1, x_0} B_{L_2, x_1} x_1 + (A_{L_1, x_0} c_{L_2} + b_{L_1}) \\ &= A_{L_1, x_0} B_{L_2} \Big[x_0^T \; \Big| \; x_1^T \Big]^T + (A_{L_1, x_0} c_{L_2} + b_{L_1}) \\ &= R_\rho(\Phi^1 \bullet \Phi^2)(x_0) \end{aligned}$$

2. Induction assumption: Let the statement be true for $L_2 \geq 2$ fixed.

3. Induction step: $L_2 \mapsto L_2 + 1$.

□

2.3.2 Network parallelization

We have shown, that there is a possiblity to simply compose networks, as is possible with functions. To further build upon that idea, we will now introduce stacking in the context of definiting a network $P(\Phi^1, \Phi^2)$ from two networks Φ^1 and Φ^2 such that $R_\rho(P(\Phi^1, \Phi^2))(x) = (R_\rho(\Phi^1)(x), R_\rho(\Phi^2)(x))$ for all $x \in \mathbb{R}^d$.

Definition 2.24 (Network Parallelization)**.** Let $L_1, L_2 \in \mathbb{N}$ s.t. $L_1 \leq L_2$ and let

$$\Phi^1 = ((A_1, b_1), \ldots, (A_{L_1}, b_{L_1})), \quad \Phi^2 = ((B_1, c_1), \ldots, (B_{L_2}, c_{L_2})),$$

be two neural networks with d-dimensional input. We call

$$P(\Phi^1, \Phi^2) := ((C_1, d_1), \ldots, (C_{L_2}, d_{L_2})),$$

the *parallelization of* Φ^1 *and* Φ^2 with

$$C_l := \left[\begin{array}{c|cc|c|cc} A_{l,x_0} & A_{l,x_1} & 0 & \cdots & A_{l,x_{l-1}} & 0 \\ B_{l,x_0} & 0 & B_{l,x_1} & \cdots & 0 & B_{l,x_{l-1}} \end{array}\right], \qquad d_l := \begin{bmatrix} b_l \\ c_l \end{bmatrix}, \qquad \text{for } 1 \leq l < L_1,$$

$$C_{L_1} := \left[\begin{array}{c|cc|c|cc|c|c|c} B_{l,x_0} & 0 & B_{l,x_1} & \cdots & 0 & B_{l,x_{L_1-1}} & B_{l,x_{L_1}} & \cdots & B_{l,x_{l-1}} \end{array}\right], \qquad d_l := c_l,$$

for $L_1 \leq l < L_2$, and

$$C_{L_2} := \left[\begin{array}{c|cc|c|cc|c|c|c} A_{L_1,x_0} & A_{L_1,x_1} & 0 & \cdots & A_{L_1,x_{L_1-1}} & 0 & 0 & \cdots & 0 \\ B_{L_2,x_0} & 0 & B_{L_2,x_1} & \cdots & 0 & B_{L_2,x_{L_1-1}} & B_{L_2,x_{L_1}} & \cdots & B_{L_2,x_{L_2-1}} \end{array}\right],$$

$$d_{L_2} := \begin{bmatrix} b_{L_1} \\ c_{L_2} \end{bmatrix}.$$

A similar construction is used for the case $L_1 > L_2$.

Lemma 2.25. In the setting of Definition (2.24), we have that $P(\Phi^1, \Phi^2)$ is a neural network with d-dimensional input, $\max\{L_1, L_2\}$ layers, $\omega_{P(\Phi^1,\Phi^2)} = \omega_{\Phi^1} + \omega_{\Phi^2}$ active weights and $N_{P(\Phi^1,\Phi^2)} = N_{\Phi^1} + N_{\Phi^2} - d$ neurons. Additionally, for all $x \in \mathbb{R}^d$, we have that

$$R_\varrho(P(\Phi^1, \Phi^2))(x) = (R_\varrho(\Phi^1)(x), R_\varrho(\Phi^2)(x)).$$

maybe include proof?

2.3.3 Standard Neural Networks

Corollary 2.26. Let ρ be the ReLU activation function. Then there are constants $C_1, C_2 > 0$ such that for any neural network Φ with d-dimensional input and $L = L(\Phi)$ layers, $N = N(\Phi)$ neurons and $M = M(\Phi)$ nonzero weights, there is a standard neural network Φ_{st} with d-dimensional input, L layers, at most $C_1 LN$ neurons and $C_2 \cdot (LN + M)$ nonzero weights such that

$$R_\rho(\Phi)(x) = R_\rho(\Phi_{\mathrm{st}})(x)$$

for all $x \in \mathbb{R}^d$.

2.3.4 Approximation Theory

For all further theoretical results we will assume that ρ, the activation function, is the ReLU as given in Definition (2.19) and furthermore define the set of functions bounded in Sobolev norm as

$$\mathcal{F}_{n,d,p,B} := \{f \in W^{n,p}((0,1)^d) \mid \|f\|_{W^{n,p}((0,1)^d)} \leq B\}.$$

To create a decent outline for this subsection, we will shortly discuss the goal and the steps to be taken; in order to show a bound on the weights and layers needed for an approximating neural network, we need to take into account the interpolation error, we may overcome with sufficient regularity. Say we would have a function f in the Sobolev space $W^{2,2}((0,1)^d)$, then it is apparent, that at least 2 derivatives of f must exists, in a piecewise, embedded sense, which can be leveraged to get a Taylor expansion around selected points.

This procedure will deduce a quantifiable error for the network approximating the initial arbitrary Sobolev function f in Sobolev norm, as long as we can bound the error, which the polynomial approximation produces with regard to the network itself.

This leaves us with two main objectives:

1. Bound the Sobolev function at critical equidistant points, using its Taylor expansion.
2. Relate a fixed error rate of the approximation polynomial and some neural network, to the networks complexity.

These two objectives are pursued by both Yarotsky (Yarotsky, 2017) and Gühring et al (Gühring et al., 2019), even though it is worth noting, that Yarotsky only provides the special case $s = 0, B = 1$, since the proof only shows an estimation with a Taylor expansion regarding the L^∞ norm, missing the crucial step of the interpolation bound to change the norm to the Sobolev expanded counterpart and additionally uses the L^∞ norm to approximate multiplication by a neural network.

Tho we will use the statement of Gühring later on, which is the extension of Yarotsky to any choice of s, p and B, we will only give the idea of the proof, since it follows the same pattern as Yarotsky's proof, but does include a lot of further theoretical setting, which will not be of much use in the actual results presented in this paper.

Proposition 2.27. There are constants $c_1, c_2, c_3 > 0$, such that for all $\varepsilon \in (0,1)$ there is a neural network $\Phi_\varepsilon^{\mathrm{sq}}$ with at most $c_1 \cdot \ln(1/\varepsilon)$ nonzero weights, at most $c_2 \cdot \ln(1/\varepsilon)$ layers, at most $c_3 \cdot \ln(1/\varepsilon)$ neurons, and with a 1-dimensional input and output such that for $f(x) := x^2$

$$\|R_\rho(\Phi_\varepsilon^{\mathrm{sq}}) - f\|_{W^{0,\infty}((0,1)^1)} \leq \varepsilon,$$

and $R_\rho(\Phi_\varepsilon^{\mathrm{sq}}(0)) = 0$.

Proof. Define the "tooth" function $g : [0,1] \to [0,1]$ with

$$g(x) = \begin{cases} 2x, & \text{for } x < 1/2, \\ 2(1-x), & \text{for } x \geq 1/2, \end{cases}$$

and its composition

$$g_s(x) := \underbrace{g \circ g \circ \cdots \circ g(x)}_{s\text{-times}}.$$

Furthermore, from the definition of the ReLU function (refer to (2.19)), we can also set $g(x) = 2\rho(x) - 4\rho(x - 1/2) + 2\rho(x - 1)$, which is intern, the same as defining a neural network with

1-dimensional input and output via $\Phi = ((A_1, b_1), (A_2, b_2))$ with

$$A_1 = \begin{bmatrix} \mathrm{Id}_{\mathbb{R}} \\ \mathrm{Id}_{\mathbb{R}} \\ \mathrm{Id}_{\mathbb{R}} \end{bmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ -1/2 \\ -1 \end{pmatrix}, \quad A_2 = \begin{bmatrix} 2 & -4 & 2 \end{bmatrix}, \quad b_2 = 0.$$

Then it is evident, using Lemma (2.23), that $g_s(x) = R_\rho(\underbrace{\Phi \odot \cdots \odot \Phi}_{s\text{-times}})(x)$ for all $x \in \mathbb{R}$ represents a network with $\mathcal{O}(s)$ layers, active weights and neurons. Now, with an explicit formulation of g_m for a fixed level $m \geq 1$, one can identify the function $f(x) := x^2$ using linear interpolation on equidistant points $k/(2^m)$, where $k = 0, \ldots, 2^m - 1$, i.e. let $\Phi_m = \Phi \odot \cdots \odot \Phi$ (m-times) be a neural network, because then for all $x \in [\frac{k}{2^m}, \frac{k+1}{2^m}]$ it is true, that

$$R_\rho(\Phi_m)(x) = \left(\frac{(k+1)^2}{2^m} - \frac{k^2}{2^m}\right)\left(x - \frac{k}{2^m}\right) + \left(\frac{k}{2^m}\right)^2.$$

Why?

Then we also know that, using the Bramble-Hilbert lemma in one-dimension (Bramble & Hilbert, 1970), the fact that $f''(x) = 2$ for all $x \in (0,1)$, and $h = 1/2^m$, that

$$\|R_\rho(\Phi_m) - f\|_{L^\infty((0,1))} \leq \max_{k=0,\ldots,2^m-1} Ch^2 \|f''\|_{L^\infty([\frac{k}{2^m}, \frac{k+1}{2^m}])} = C' 2^{-2m}.$$

Hence setting $m = \lceil \ln(1/\varepsilon) \rceil$, with $C, C' > 0$ constants, only dependent on the domain, we achieve the wanted bound from the statement for the neural network $\Phi_\varepsilon^{\mathrm{sq}} := \Phi_m$ with $\mathcal{O}(m) = \mathcal{O}(\ln(1/\varepsilon) + 1) \leq \mathcal{O}(2\ln(1/\varepsilon))$ layers, neurons and active weights. □

This statement enables us to express multiplication using a neural network. This is straight forward, when viewing the polarization identity:

$$xy = \frac{1}{2}((x+y)^2 - x^2 - y^2), \quad \text{for } x, y \in \mathbb{R}, \tag{2.1}$$

as given by the first binomial formula.

Proposition 2.28. Let $M > 0$ and $\varepsilon \in (0,1)$, then there is a ReLU neural network $\widetilde{\times}$ with 2-dimensional input and 1-dimensional output, and constants $c_1 > 0$ and $c_2 = c_2(M) > 0$, such that for $\times : \mathbb{R} \times \mathbb{R} \to \mathbb{R}; (x,y) \mapsto xy$

1. $\|R_\rho(\widetilde{\times}) - \times\|_{W^{0,\infty}((-M,M)^2)} \leq \varepsilon$,
2. if $x = 0$ or $y = 0$, then $R_\rho(\widetilde{\times})(x,y) = 0$,
3. the number of layers, active weights and neurons of $\widetilde{\times}$ is at most $c_1 \ln(1/\varepsilon) + c_2$.

Proof. Let $\delta := \varepsilon/(6M^2)$. Then we construct an approximation of the square function $f(x) = x^2$ for $x \in (0,1)$ using Proposition (2.27) for the chosen δ, i.e. we define $\Phi_\delta^{\mathrm{sq}}$, such that

$$\|R_\rho(\Phi_\delta^{\mathrm{sq}}) - f\|_{W^{0,\infty}((0,1)^1)} \leq \frac{\varepsilon}{6M^2},$$

which now of course has $\mathcal{O}(\ln(1/\delta))$ many layers, active weights and neurons. Since the statement necessarily needs to implement negative numbers as well, we show with $|x| = \rho(x) + \rho(-x)$ that it is indeed possible to expand any neural network with a positive domain to a negative domain using a neural network (by simply concatenating). Thus we construct $\widetilde{\times}$ via adding an addition layers on top of parallel square networks with scaled input, achieving

$$R_\rho(\widetilde{\times})(x,y) = 2M^2 \left(R_\rho(\Phi_\delta^{\mathrm{sq}}) \left(\frac{|x+y|}{2M} \right) - R_\rho(\Phi_\delta^{\mathrm{sq}}) \left(\frac{|x|}{2M} \right) - R_\rho(\Phi_\delta^{\mathrm{sq}}) \left(\frac{|y|}{2M} \right) \right).$$

To appreciate the third statement of our Proposition, we may now put together the necessary complexity of this network using the constants in Proposition (2.27);

1. We have 2 layers for the implementation of the absolute value, 1 for the addition in the end, and $c_0 \ln(1/\delta)$ for the square function, thus $L_{\widetilde{\times}} \leq c_0 \ln(1/\varepsilon) + c_0 \ln(6M^2) + 3$.
2. We need $4+2+2$ active weights for the first matrix to distribute each (x,y) onto the parallel networks together with 6 active weights for the superposing of the input into the $|\cdot|$-function. Another 3 are required for the last layer adding all outputs. Of course each of the different computiations for the artificial scaled square function must be in parallel, thus $\omega_{\widetilde{\times}} \leq 3c_0 \ln(1/\varepsilon) + 3c_0 \ln(6M^2) + 17$.
3. A similar consideration gives us $9 = (2+1) \cdot 3$ neurons for the first two and the last layer, thus giving us $N_{\widetilde{\times}} \leq 3c_0 \ln(1/\varepsilon) + 3c_0 \ln(6M^2) + 9$.

Finally taking the adequate maximum, and remarking, that c_0 does not depend on M, we get the wanted bound in (2).

To make the true multiplication match the form of our implemented network, using (2.1), we may rewrite for $x, y \in (-M, M)$

$$\begin{aligned} xy &= 4M^2 \left(\frac{x}{2M} \cdot \frac{y}{2M} \right) \\ &= 4M^2 \frac{1}{2} \left(\left(\frac{x}{2M} + \frac{y}{2M} \right)^2 - \left(\frac{x^2}{2M} \right)^2 \left(\frac{y^2}{2M} \right)^2 \right) \\ &= 2M^2 \left(\left(\frac{|x+y|}{2M} \right)^2 - \left(\frac{|x|}{2M} \right)^2 - \left(\frac{|y|}{2M} \right)^2 \right). \end{aligned}$$

Now with the deviation from each of the square approximation we get,

$$\begin{aligned} \| R_\rho(\widetilde{\times}) - \times \|_{L^\infty((0,1)^2)} &\leq 2M^2 \left(3 \cdot \sup_{\frac{|x|}{2M} : x \in (-M,M)} |\, R_\rho(\Phi_\delta^{\mathrm{sq}})(\widetilde{x}) - \widetilde{x}^2 \,| \right) \\ &\leq 6M^2 \delta = \varepsilon. \end{aligned}$$

We note a few things happening here for clarity; it is possible to simply use the reverse triangle equality to get all three term errors added, and then put them together. This can be done, since the $L^\infty((0,1)^2)$ norm appreciates only the maximal value of the error across all possible elements of the $(0,1)^2$ domain, but because the inner function is bijective, these are just the same. Additionally we use the, trivial by definition, statement

$$\|g \circ f\|_{L^\infty((0,1))} \leq \|g\|_{L^\infty((0,1))}.$$

This has shown the statement (3).

Statement (1) on the other hand, is a direct consequence of the subliminally result in Proposition (2.27). $\square$

As a last step, before stating and proving the Yarotsky Theorem, we want to discuss the construction and complexity of a network approximating any collection of equidistantly spaced function with local support in a higher dimensional domain $(0,1)^d$ for $1 \leq d < \infty$. If we take into account, that we may also want these functions to form a partition of unity on the high dimensional domain, this will be ground to tackle the complexity of the approximation of polynomials with neural networks.

Lemma 2.29. For any $d, N \in \mathbb{N}$ there is a collection of functions

$$\Psi = \left\{ \phi_m \,|\, m \in \{0, \dots, N\}^d \right\},$$

with $\phi : \mathbb{R}^d \to \mathbb{R}$ for all $m = (m_1, \dots, m_d) \in \{0, \dots, N\}^d$ with the following properties:

1. $0 \leq \phi_m(x) \leq 1$ for all $\phi_m \in \Psi$ and every $x \in \mathbb{R}^d$,
2. $\sum_{\phi_m \in \Psi} \phi_m(x) = 1$ for every $x \in [0,1]^d$,
3. $\mathrm{supp}(\phi_m) \subset B_{1/N, \|\cdot\|_{\ell^\infty}}(m/N)$ for all $\phi_m \in \Psi$,
4. there are constants $C, c \geq 1$ such that for each $\phi_m \in \Psi$ there is a neural network Φ_m with d-dimensional input and output, at most 3 layers, $C \cdot d$ active weights and neurons, such that

$$\prod_{l=1}^{d} [R_\rho(\Phi_m)]_l = \phi_m$$

and $\|[R_\rho(\Phi_m)]_l\|_{L^\infty((0,1)^d)} \leq 1$ for all $l = 1, \dots, d$.

Proof. The idea of the proof, will be one of construction, where we will find adequate functions, which are easily transferable into ReLU neural networks. For this, define the functions

$$\psi : \mathbb{R} \to \mathbb{R}, \quad \psi(x) = \begin{cases} 1, & |x| < 1, \\ 0, & 2 < |x|, \\ 2 - |x| & 1 \leq |x| \leq 2, \end{cases}$$

and $\phi_m : \mathbb{R}^d \to \mathbb{R}$ as the prodoct of shifted and scaled versions of ψ, which inherit its local support, i.e.

$$\phi_m(x) := \prod_{l=1}^{d} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right),$$

for $m = (m_1, \dots, m_d) \in \{0, \dots, N\}^d$. This function satisfies our first three parts of the Lemma directly: Take one x_i for some $i = 1, \dots, d$. Now we get that $\psi(3N(x_i - (m_i/N))) \in [-2, 2]$, since ψ has support of $[-2, 2]$. This gives us

$$x \in \left[\frac{m}{N} - \frac{2}{3N}, \frac{m}{N} + \frac{2}{3N}\right],$$

yielding the third property, as we took i arbitrarily. Simultaneously we use the zero product property, and the fact, that the range of ψ is $[0,1]$ to get the first property. Defining $z_i := 3Nx_i$, we can rewrite for any N the one-dimensional function

$$S(x_i) := \sum_{m_i=0}^{N} \psi\left(3N\left(x_i - \frac{m_i}{N}\right)\right) \quad \text{into the function} \quad \widetilde{S}(z_i) := \sum_{m_i=0}^{N} \psi(z_i - 3m_i),$$

that intern gives us the wanted result of $\widetilde{S}(z_i) = 1$ for any $z_i \in [0, 3N]$, because the least distance between points is 3, which means going forwards or backwards either lands one outside the support of the function, if $z_i \in [-1, 1]$, or it adds both third case values, i.e. setting $z_i \in [1, 2]$ w.l.o.g

$$\psi(z_i) + \psi(z_i - 3) = (2 - z_1) + (2 + (z_i - 3)) = 1.$$

Now we can extrapolate this result to S by a simple scaling argument and then use Fubini and the factorization of the sum over products to get

$$\sum_{m \in \{0,\dots,N\}^d} \phi_m(x) = \sum_{m_1=0}^{N} \cdots \sum_{m_d=0}^{N} \prod_{l=1}^{d} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right) = \prod_{l=1}^{d}\left(\sum_{m_l=0}^{N} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right)\right) = 1.$$

Finally we are only left with proving the forth property. For this, we need to construct a neural network Φ_ψ, that realizes ψ. Let

$$A_1 := \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad b_1 := \begin{pmatrix} 2 \\ 1 \\ -1 \\ -2 \end{pmatrix}, \quad A_2 := \left[0 \,\middle|\, 1 \quad -1 \quad -1 \quad 1\right], \quad b_2 := 0,$$

define the neural network $\Phi_\psi = ((A_1, b_1), (A_2, b_2))$. With distinction by cases one can easily see, that for all $x \in \mathbb{R}$

$$\begin{aligned} R_\rho(\Phi_\psi)(x) &= A_2(\rho(A_1 x + b_1)) + b_2 \\ &= A_2\left(\begin{bmatrix}\rho(x+2) & \rho(x+1) & \rho(x-1) & \rho(x-2)\end{bmatrix}^T\right) \\ &= \rho(x+2) + \rho(x-2) - (\rho(x+1) + \rho(x-1)) \\ &= \psi(x). \end{aligned}$$

Furthermore the complexity of this network is 2 layers, 12 active weights and 6 neurons. Let $\Phi_{m,l}$ be a neural network with d-dimensional input and one-dimensional output, such that

$$R_\rho(\Phi_{m,l})(x) = 3N\left(x_l - \frac{m_l}{N}\right), \quad \text{for all } x \in \mathbb{R}^d \text{ and } m \in \{0, \dots, N\}^d.$$

Then the complexity of this network is 1 layer, 2 active weights and $d + 1$ neurons. Finally realizing that the composition of both these functions, i.e. the realization of the concatenation of networks, would yield each factor for the wanted function. If we now parallelize this concatenation, we get the forth property by definition, i.e. let

$$\Phi_m := P(\Phi_\psi \odot \Phi_{m,l} \,|\, l = 1, \dots, d),$$

since then

$$\prod_{l=1}^{d} [R_\rho(\Phi_m)]_l(x) = \phi_m(x), \quad \text{for all } x \in \mathbb{R}^d.$$

This network now has d-dimensional input and output and the complexity, using both Lemma (2.23) and Lemma (2.25), of 3 layers, at most $(12 + 2) \cdot 2 \cdot d = 28d$ active weights, and at most $9d$ neurons. □

With these auxiliary results, we may finally start with the statement and proof of the Yarotsky Theorem.

Theorem 2.30 (Yarotsky Theorem). *Let $d \in \mathbb{N}, n \in \mathbb{N}, B = 1$ and $f \in \mathcal{F}_{n,d,\infty,B}$ some arbitrary function. Then there is a constant $c = c(d,n) > 0$ with the following properties:*

For any $\varepsilon \in (0,1)$, there is a neural network architecture $\mathcal{A}_\varepsilon = \mathcal{A}_\varepsilon(d,n,\varepsilon)$ with d-dimensional input and one-dimensional output such that, there is a neural network Φ_ε^f that has architecture $\mathcal{A}_\varepsilon$ and satisfies

$$\|f - R_\rho(\Phi_\varepsilon^f)\|_{W^{0,\infty}((0,1)^d)} \leq \varepsilon,$$

and

1. $L_{\mathcal{A}_\varepsilon} \leq c \cdot \ln(1/\varepsilon)$,
2. $\omega_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/n} \ln(1/\varepsilon)$,
3. $N_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/n} \ln(1/\varepsilon)$.

Proof. Given the definition of the functions $\phi_m \in \Psi$ as constructed in the proof of Lemma (2.29), we want to structure the proof of this Theorem in two sub-goals:

The formulation of an approximate function f_1 for f using the partition of unity and an adequate Taylor expansion, and the subsequent approximation of that function f_1 with a neural network Φ_{f_1} using the $\widetilde{\times}$-multiplication of Proposition (2.28).

Step 1: Taylor Expansion

For any $m \in \{0, \dots, N\}^d$, consider the $(n-1)$-Taylor polynomial for the function f at $x_0 = m/N$:

$$P_m(x) = \sum_{n \colon |n| < n} \frac{D^n f(x)}{n!}\bigg|_{x=\frac{m}{N}} \left(x - \frac{m}{N}\right)^n, \quad \text{for } x \in \mathbb{R}^d,$$

with the conventions $n! = \prod_{l=1}^d n_l!$ and $(x - m/N)^n = \prod_{l=1}^d (x_l - \frac{m_l}{N})^{n_l}$. With the partition of unity of Lemma (2.29), we can then set any function, as a linear combination of its point-values and the partition function. Especially this allows us to define

$$f = \sum_{m \in \{0,\dots,N\}^d} \phi_m f, \quad \text{and} \quad f_1 = \sum_{m \in \{0,\dots,N\}^d} \phi_m P_m.$$

Now we can exploit the properties given by Lemma (2.29), as follows; firstly use the $\|\phi_m\|_{L_\infty((0,1)^d)} = 1$ property in Line (2.3), then one can estimate the maximal number of partitions with 2^d in Line (2.4). After this we take the simple Lagrange remainder

argue this

in Line (2.5). Lastly one can simply argue with $f \in \mathcal{F}_{n,d,\infty,B=1}$ in Line (2.6).

$$\begin{aligned}
\|f - f_1\|_{L_\infty((0,1)^d)} &\leq \sum_{m \in \{0,\dots,N\}^d} \|\phi_m (f - P_m)\|_{L_\infty((0,1)^d)} && (2.2)\\
&\leq \sum_{m: \left|x_l - \frac{m_l}{N}\right| < \frac{1}{N}, \forall l=1,\dots,d} \|f - P_m\|_{L_\infty((0,1)^d)} && (2.3)\\
&\leq 2^d \max_{m: \left|x_l - \frac{m_l}{N}\right| < \frac{1}{N}, \forall l=1,\dots,d} \|f - P_m\|_{L_\infty((0,1)^d)} && (2.4)\\
&\leq \frac{2^d d^n}{n!} \left(\frac{1}{N}\right)^n \max_{n:\, |n|=n} \|D^n f\|_{L_\infty((0,1)^d)} && (2.5)\\
&\leq \frac{2^d d^n}{n!} \left(\frac{1}{N}\right)^n && (2.6)
\end{aligned}$$

Now taking

$$N = \left\lceil \left(\frac{n!}{2^d d^n} \cdot \frac{\varepsilon}{2} \right)^{-\frac{1}{n}} \right\rceil$$

gives the approximation bound of

$$\|f - f_1\|_{L_\infty((0,1)^d)} \leq \frac{\varepsilon}{2}. \tag{2.7}$$

Step 2: Approximation by a Neural Network

The polynomial function f_1 may be further expressed in a different form, which allows us to seperately compare it to a potential network approximation using the definition of ϕ_m, i.e. for $x \in \mathbb{R}^d$

$$f_1(x) = \sum_{m \in \{0,\dots,N\}^d} \sum_{n:\, |n|<n} a_{m,n} \underbrace{\left(\phi_m(x) \left(x - \frac{m}{N} \right)^n \right)}_{=: f_{m,n}(x)}. \tag{2.8}$$

This expansion now has at most $d^n(N+1)^d$ terms containing $f_{m,n}$, since

$$\left| \left\{ n \in \{0, \dots, n-1\}^d \,:\, |n| < n \right\} \right| \leq \left| \left\{ n \in \{0, \dots, n-1\}^d \right\} \right| = d^n,$$

and obviously $|\{m = \{0,\dots,N\}^d\}| = (N+1)^d$. We note, that we take $M = d + n$ for the definition of the network multplication, i.e. $\widetilde{\times} : (-M, M)^2 \to \mathbb{R}$. This is done, because we have as an obvious consequence from the fact, that if $x, y \in (-1,1)$, that

$$|\, R_\rho(\widetilde{\times})(x,y) - xy \,| \leq \delta \quad \Longrightarrow \quad |\, R_\rho(\widetilde{\times})(x,y) \,| \leq |y| + \delta,$$

for some fixed but not yet specified $\delta \in (0,1)$. Recursively concatenating $\widetilde{\times}$ for $(d+n-1)$-times can then propagate the error to at most $M = (d+n)$ upon the last realization of $\widetilde{\times}$.

As aforementioned, we can now compare each function $f_{m,n}$ pointwise to a realization of a network approximating at most $n-1$ of the multiplications needed for $\prod_{l=1}^d (x_l - (m_l/N))^{n_l}$, as we know that $\sum_{l=1}^d n_l < n$, d additional multiplications for the realization of the function ϕ_m and one multiplication for putting it together. Hence we define a neural network $\Phi_{f_1}^{m,n}$ with d-dimensional input and one dimensional output via

$$
\begin{aligned}
R_\rho(\Phi_{f_1}^{m,n})(x) = R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_1(x_1), R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_2(x_2), \dots \\
R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_d(x_d), R_\rho(\widetilde{\times})\Big(x_1 - \frac{m_1}{N}, \dots \\
R_\rho(\widetilde{\times})\Big(x_{d-1} - \frac{m_{d-1}}{N}, R_\rho(\widetilde{\times})\Big(x_d - \frac{m_d}{N}, \dots \\
R_\rho(\widetilde{\times})\Big(x_d - \frac{m_d}{N}, x_d - \frac{m_d}{N}\Big)\Big)\Big)\Big)\Big)\Big)\Big).
\end{aligned}
$$

for $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ where we used Φ_m as in the proof Lemma (2.29). This network contains at most $(d+n)$ concatenations of $\widetilde{\times}$, and thus by Lemma (2.29) and its constant $C > 0$, Lemma (2.23), and Proposition (2.28) and its constants $c_0, c_1(d,n) > 0$,

$$
L_{\Phi_{f_1}^{m,n}} \leq 3c_0(d+n)\left(\ln\left(\frac{1}{\delta}\right) + (d+n)c_1\right), \quad \omega_{\Phi_{f_1}^{m,n}} \leq C2(d+n)d\left(\ln\left(\frac{1}{\delta}\right) + 2(d+n)C\right). \tag{2.9}
$$

for a given precision level $\delta \in (0,1)$, where the number of neurons are of the same order as the number of active weights.

The error comparison of $R_\rho(\Phi_{f_1}^{m,n})$ and $f_{m,n}$ can then be done by counting the maximal depth of usages of the network $\widetilde{\times}$, which is $(d+n)$ as shown above, and the estimate of

$$
\left|\left(x - \frac{m}{N}\right)\right| \leq 1, \quad |\,\psi(x)\,| \leq 1,
$$

for all $x \in (0,1)^d$. Then directly

$$
\|R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n}\|_{L_\infty((0,1)^d)} \leq (d+n)\delta.
$$

Now since, as mentioned below Equation (2.8), f_1 has at most $d^n(N+1)^d$ terms, but since any $x \in \mathbb{R}^d$ can be only in the support of at most 2^d functions ϕ_m by the second statement in Proposition (2.28), the sum going over all m can be reduced to only those, which are non-zero, which we employ in Line (2.11). This brings $(N+1)^d \to 2^d$, supporting the equation in Line (2.12). Setting

$$
\delta = \frac{\varepsilon}{2 \cdot 2^d d^n (d+n)}
$$

allows us to finally assert, using the bound of Equation (2.9) for Line (2.13),

$$
\begin{aligned}
\|R_\rho(\Phi_{f_1}) - f_1\|_{L_\infty((0,1)^d)} &\leq \sum_{m \in \{0,\dots,N\}^d} \sum_{n\colon |n|<n} |a_{m,n}|\,\|R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n}\|_{L_\infty((0,1)^d)} &(2.10)\\
&= \sum_{m\colon x \in \operatorname{supp}(\phi_m)} \sum_{n\colon |n|<n} |a_{m,n}|\,\|R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n}\|_{L_\infty((0,1)^d)} &(2.11)\\
&\leq 2^d d^n \max_{m\colon x \in \operatorname{supp}(\phi_m)} \max_{n\colon |n|<n} \|R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n}\|_{L_\infty((0,1)^d)} &(2.12)\\
&\leq 2^d d^n (d+n)\delta &(2.13)\\
&= \frac{\varepsilon}{2}. &(2.14)
\end{aligned}
$$

Our knowledge of δ together with at most $d^n(N+1)^d = \mathcal{O}(\varepsilon^{-d/n})$ parallel networks with one additive layer and the fact that generally we have $c' \ln(1/\delta + c'') \leq c''' \ln(1/\delta)$ for $c', c'' > 0$ and $c''' \geq c' \ln(1/c'')$, gets us the final complexity of Φ_{f_1} from the starting point of Equation (2.9), i.e.

$$L_{\Phi_{f_1}} \leq c \ln(1/\varepsilon), \quad \omega_{\Phi_{f_1}} \leq c\varepsilon^{-d/n} \ln(1/\varepsilon), \quad N_{\Phi_{f_1}} \leq c\varepsilon^{-d/n} \ln(1/\varepsilon).$$

with some constant $c = c(d, n) > 0$.

Using the triangle inequality, we can show that this network indeed satisfies the wanted precision bound with both errors stemming from Equation () and Equation () respectively:

$$\|f - R_\rho(\Phi_{f_1})\|_{L_\infty((0,1)^d)} \leq \|f - f_1\|_{L_\infty((0,1)^d)} + \|f_1 - R_\rho(\Phi_{f_1})\|_{L_\infty((0,1)^d)} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

We may even expand Yarotsky to cover a more general case, where $f \in \mathcal{F}_{n,d,p,B}$ for arbitrarily $1 \leq p \leq \infty$ and $B > 0$ and the norm, to which we assert the precision, an arbitray Sobolev norm. This extension is presented by Gühring et al. (Gühring et al., 2019).

Theorem 2.31 (Gühring Theorem)**.** *Let $d \in \mathbb{N}, n \in \mathbb{N}_{\geq 2}, B > 0$ and $0 \leq s \leq 1$ and $f \in \mathcal{F}_{n,d,p,B}$ some arbitrary function. Then there is a constant $c = c(d, n, p, B, s) > 0$ with the following properties:*

For any $\varepsilon \in (0, 1)$, there is a neural network architecture $\mathcal{A}_\varepsilon = \mathcal{A}_\varepsilon(d, n, p, B, s, \varepsilon)$ with d-dimensional input and one-dimensional output such that, there is a neural network Φ_ε^f that has architecture $\mathcal{A}_\varepsilon$ and satisfies

$$\|f - R_\rho(\Phi_\varepsilon^f)\|_{W^{s,p}((0,1)^d)} \leq \varepsilon,$$

and

1. $L_{\mathcal{A}_\varepsilon} \leq c \cdot \ln(1/\varepsilon)$,
2. $\omega_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/(n-s)} \ln(1/\varepsilon)$,
3. $N_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/(n-s)} \ln(1/\varepsilon)$.

Proof Sketch. Considering that all of the tools used for this proof will neither be of any use to the main results in this paper nor to the general structure, we will only sketch the general steps necessary to achieve it for the interested reader:

1. The first step is to expand the Proposition (2.27) to the $W^{1,\infty}(0,1)$-norm. This is done by simple calculation, since an explicit construction is being made.

2. This can be applied to Proposition (2.28) in the same way. Additionally we need to bound $|\, R_\rho(\widetilde{\times}) \,|_{W^{1,\infty}}((0,1)^2)$, since this is required in for the bound on the collective error under the Taylor expanded function in the end.

3. The most import additional step is to bound the switch of norms from $W^{1,p}((0,1)^d)$ to $W^{s,p}((0,1)^d)$ of a function being approximated by the partition of unity, that we discussed in Lemma (2.29). This gives

$$\|f - f_N\|_{W^{s,p}((0,1)^d)} \leq C \left(\frac{1}{N}\right)^{n-s} \|f\|_{W^{n,p}((0,1)^d)},$$

for a partition of unity of size N. It is worth noting, that this factor in front is responsible for the different approximation in regard to weights and neurons, since it influences the necessary choice of δ.

4. The last step, which differs greatly from the proof of Yarotsky Theorem (2.30), is the approximation of a sum of localized polynomials with a neural network. Here the challenge lies again in the control of the derivatives.

□

2.3.5 Statistical Learning

We have now introduced the framework for analytical discussion regarding complexity and potential of neural networks. What is clearly missing however, is the practical "achievement" of a true neural network Φ fitting the function in question from some architecture $\mathcal{A}_\Phi$, that a machine learning engineer thought of. This crucial step in the process of development is refered to as *learning*. This subsection is my own work, if not cited otherwise.

In order to quantify the "falseness" of a potential attribution, we need a function, that compares a true solution to an output of the network operating with the true parameters connected to the true solution. We remark that the following two definitions are based upon the paper (Ciampiconi et al., 2024), where the interested reader can find extensive terminology and currently used loss functions and their optimization techniques.

Definition 2.32 (Feature and Label Space)**.** Let $\mathcal{X}$ be the so-called *feature space* and $\mathcal{Y}$ the *label space* of the underlying unknown function

$$f : \mathcal{X} \to \mathcal{Y},$$

classifying each feature with a label.

The general goal of all machine learning problems is to closely approximate that classification function f.

Definition 2.33 (Loss Function)**.** Given $f_\theta : \mathcal{X} \to \mathcal{Y}$ arbitrarily and a feature and label space as in Definition (2.32) and a sample $\{x_i, y_i\}_{i=1}^N \subset \mathcal{X} \times \mathcal{Y}$ for some $N \in \mathbb{N}$, we define the general loss function as a mapping of $f(x_i)$ with its corresponding y_i to a non-negative real number, i.e.

$$\mathcal{L}(f_\theta | \{x_1, \ldots, x_N\}, \{y_1, \ldots, y_N\}) \in \mathbb{R}^+,$$

such that for f being the function from Definition (2.32) for all $N \in \mathbb{N}$

$$\mathcal{L}(f | \{x_1, \ldots, x_N\}, \{y_1, \ldots, y_N\}) = 0.$$

Definition 2.34 (Neural Network Weights)**.** Suppose for some $d, L \in \mathbb{N}$ we have a neural network architecture $\mathcal{A}_\Phi$ of input dimension d, L layers and output dimension N_L according to Definition (2.15) and Φ be a neural network of that architecture (see Definition (2.18)). Further let $f : \mathbb{R}^d \to \mathbb{R}^{N_L}$ be some arbitrary function. Define the *masks of* $\mathcal{A}_\Phi$ via

$$I_l := \{(i,j) \,:\, [A_l]_{ij} \neq 0\}, \quad J_l := \{i \,:\, [b_l]_i \neq 0\},$$

for each layer $1 \leq l \leq L$. We further set the *admissible weight space* as

$$\Theta(\mathcal{A}_\Phi) := \prod_{l=1}^{L} \left(\mathbb{R}^{I_l} \times \mathbb{R}^{J_l}\right),$$

and its elements $\theta \in \Theta(\mathcal{A}_\Phi)$ with

$$\theta = \left((\theta^A_{l,ij})_{(i,j)\in I_l}, (\theta^b_{l,i})_{i\in J_l} \right)_{l=1}^{L}.$$

Then in each layer of $\Phi(\theta) = ((\hat{A}_1(\theta), \hat{b}_1(\theta)), \ldots, (\hat{A}_L(\theta), \hat{b}_L(\theta)))$ the affine transformation can be uniquely retrieved via the lifting

$$\hat{A}_l(\theta)_{ij} = \begin{cases} \theta^A_{l,ij}, & (i,j) \in I_l, \\ 0, & \text{otherwise}, \end{cases} \qquad \hat{b}_l(\theta)_i = \begin{cases} \theta^b_{l,i}, & i \in J_l, \\ 0, & \text{otherwise}. \end{cases}$$

Then the network Φ will be called a *fitting network with regard to* f, if $\Phi := \Phi(\theta^*)$ is given such that

$$\theta^* = \underset{\theta \in \Theta(\mathcal{A}_\Phi)}{\arg\min}\, \mathcal{L}\left(R_\rho(\Phi(\theta)) | \{x_1, \ldots, x_N\}, \{f(x_1), \ldots, f(x_N)\}\right)$$

for some sample $\{x_1, \ldots, x_N\}_{i=1}^N \subset \mathbb{R}^d$ with $N \in \mathbb{N}$ and a suitable choice of a loss function $\mathcal{L}$. Note, that this minimizing condition and the architecture are what, not necessarily uniquely, determines the fitting network.

Chapter 3

Formulation of the General Problem

Definitely add traditional convergence and computational cost.

Let $\Theta \subset \mathbb{R}^p$ and $\Theta' \subset \mathbb{R}^q$ be compact and denote the geometric and physical parameter spaces respectively. Further we let $\Omega \subset \mathbb{R}^d$ be some Lipschitz domain and $[0, T]$ be a time interval for some $T > 0$ of a parametric time-dependent PDE, i.e. for any $\mu \in \Theta$ and $\nu \in \Theta'$

$$u^{\mu,\nu} := u(\mu, \nu), \quad \text{in a sufficiently smooth space,}$$

is the (classical) solution of the PDE, for the choice of the space depending on the given specified problem, if

$$\begin{cases} \dfrac{\partial}{\partial t} u^{\mu,\nu}(x,t) + \mathcal{L}(\mu,\nu) u^{\mu,\nu}(x,t) + \mathcal{N}(u^{\mu,\nu}(x,t),\mu,\nu) = f(x,t;\mu,\nu), & (x,t) \in \Omega \times [0,T], \\ \mathcal{B}(\mu,\nu) u^{\mu,\nu}(x,t) = g_N(x,t;\mu,\nu) & (x,t) \in \partial\Omega \times [0,T], \\ u^{\mu,\nu}(x,t) = u_0^{\mu,\nu}(x) & (x,t) \in \Omega \times \{0\}, \end{cases}$$

where $\mathcal{L}$ represents a linear, and $\mathcal{N}$ represents a non-linear integro-differential operator. Using calculus of variation or distribution theory, one can then formulate a weak form of the problem and find a general space, most commonly Sobolev (for elliptic problems with coersive properties), Bochner (for parabolic problems with coersive properties on the stationary part) or discontinuous Galerkin spaces (for hyperbolic problems with enough regularity), on which a weak solution can be defined. Such a space $\mathcal{H}$ is generally some Banach space, most often a subspace of L^2.

This weak solution will have less regularity imposed than the strong formulation, but this sometimes allows us to retrieve existence and uniqueness results. Quite famously, there are counter examples of course; the Navier-Stokes Equation in three dimensions poses one such result, where a weak form exists and numerical methods have been developed, which can provide a solution in special cases, but existence remains unproven in general. As mentioned beforehand, we will completely omit the exact transformation of the strong PDE formulation into a weak one, and simple assume that this can be done:

Say we (can) discretize the problem in the spacial domain Ω with N_h degrees of freedom, i.e. let $\mathcal{H}_h \subset \mathcal{H}$ be a Hilbert space with $\mathcal{H}_h \cong \mathbb{R}^{N_h}$, a scalar product $\langle \cdot, \cdot \rangle_{\mathcal{H}_h}$ and a corresponding norm $\|\cdot\|_{\mathcal{H}_h}$, such that the resulting full order model solution can be identified using the formulation

$$t \mapsto u_h(\cdot, t; \mu, \nu) = \sum_{j=1}^{N_h} [u_h^{\mu,\nu}(t)]_j \, \varphi_j(\mu) \in C^1([0,T]; \mathcal{H}_h) \tag{3.1}$$

with $\{\varphi_j(\mu)\}_{j=1}^{N_h}$ a basis of $\mathcal{H}_h$ for each $\mu \in \Theta$ and $u_h^{\mu,\nu}(t) \in \mathbb{R}^{N_h}$ for each $t \in [0,T]$, and the underlying problem for the coefficients

$$\begin{cases} M(\mu)\dfrac{\partial}{\partial t}u_h^{\mu,\nu}(t) + A(\mu,\nu)u_h^{\mu,\nu}(t) + N(u_h^{\mu,\nu}(t),\mu,\nu) = f(t;\mu,\nu) & t \in [0,T], \\ u_h^{\mu,\nu}(0) = u_0(\mu,\nu) & t = 0, \end{cases} \tag{3.2}$$

where we assume

- $M : \Theta \to \mathbb{R}^{N_h \times N_h}$ to send the geometrical parameter to a mass matrix,
- $A : \Theta \times \Theta' \to \mathbb{R}^{N_h \times N_h}$ to send the set to a symmetric positive definite stiffness matrix,
- $N : \mathbb{R}^{N_h} \times \Theta \times \Theta' \to \mathbb{R}^{N_h}$ to be some non-linear term,
- $f : [0,T] \times \mathbb{R}^{N_h} \times \Theta \times \Theta' \to \mathbb{R}^{N_h}$ to be the corresponding source term of the equation.

Additionally let $u_0 : \Theta \times \Theta' \to \mathbb{R}^{N_h}$ be the inital data. The mass matrix is defined for each $\mu \in \Theta$ as follows

$$M(\mu) = \begin{bmatrix} \langle\varphi_1(\mu),\varphi_1(\mu)\rangle_{\mathcal{H}_h} & \cdots & \langle\varphi_1(\mu),\varphi_{N_h}(\mu)\rangle_{\mathcal{H}_h} \\ \vdots & \ddots & \vdots \\ \langle\varphi_{N_h}(\mu),\varphi_1(\mu)\rangle_{\mathcal{H}_h} & \cdots & \langle\varphi_{N_h}(\mu),\varphi_{N_h}(\mu)\rangle_{\mathcal{H}_h} \end{bmatrix}.$$

This allows us to equip $\mathbb{R}^{N_h}$, the space of the coefficients of any solution in $\mathcal{H}_h$, with the following norm

$$\|u_h^{\mu,\nu,t}\| := \sqrt{(u_h^{\mu,\nu,t})^T\, M(\mu)\, u_h^{\mu,\nu,t}}, \tag{3.3}$$

which is, in fact, the equivalent of the norm above, i.e. $\|u_h^{\mu,\nu,t}\| = \|u_h(\cdot,t;\mu,\nu)\|_{\mathcal{H}_h}$.

Introducing the mass matrix norm will make all data driven procedures somewhat intrusive, since the norm will depend on mu and nu, maybe not even a problem, since the choice needs to be done a priori, but what about pod and projections

Note that it is indeed possible to arrive at that discretization without skipping to a weak formulation, e.g. using finite differences or finite volume schemes, but since any existence statement needs a more general space than C^n to guarantee completeness and hence imposes some identity using a basis, we need only to infer such a procedure. For the interested reader, an explicit discretization using finite volume schemes can be seen in (Haasdonk & Ohlberger, 2008).

This formulation establishes smoothness with regard to time, however we will implicitly further discretize in time using $\{t_k\}_{k=1}^{N_t} := \mathcal{T}$ time steps by gathering data only at these points. This is called *method of lines approach* (Hesthaven et al., 2022, p. 278).

Definition 3.1 (Solution Data)**.** Following this, we may denote with

$$u_h^{\mu,\nu,t} := u_h^{\mu,\nu}(t) \in \mathbb{R}^{N_h}, \quad \text{for } \mu \in \Theta, \nu \in \Theta' \text{ and all } t \in \mathcal{T},$$

a finite dimensional *solution snapshot* and

$$u_h^{\mu,\nu} := [u_h^{\mu,\nu}(t)]_{t\in\mathcal{T}} \in \mathbb{R}^{N_h \times N_t}, \quad \text{for } \mu \in \Theta \text{ and } \nu \in \Theta',$$

with $N_t := |\mathcal{T}|$ a finite dimensional *solution trajectory* of the discretized parametric PDE. Furthermore, let $\mathcal{P}_1 \subset \Theta$, $\mathcal{P}_2 \subset \Theta'$ and $\mathcal{P} = \mathcal{P}_1 \times \mathcal{P}_2$ be some finite parameter data with respective sizes $N_{s_1}, N_{s_2} \in \mathbb{N} \cup \{\infty\}$ and $N_s = N_{s_1}N_{s_2}$.

Mainly, we stated the problem framework, to know with which class of of PDEs we will be working with. Hence any regularity assumption is only given here in order to assert an outlook on the use of L^p norms for the error between the weak solution $u \in \mathcal{H}$, the finite dimenisonal solution $u_h \in \mathcal{H}_h$ and the approximation of that solution $\hat{u}_h \in \mathcal{H}_h$ (of reduced order) using ROM methods.

It is the latter ones, we will be concerned with in this thesis, which are however clearly embedded in L^2, since they are finite dimensional and bounded by an upcoming assumption.

From this point forward, we want to consistenly rely on the following few assumptions, which regulate the discrete problem above. The first Assumption (1) justifies a nuanced approach regarding the geometric parameter μ and the physical parameter ν. The second Assumption (2) is necessary for the derivation of any convergence result to a true parameter-to-solution map, when sampling an ever growing number of examples, while the third Assumption (3) imposes regularity on the mentioned parameter-to-solution map. Interestingly enough, it is that Assumption, which ensures the existence of a true solution of the discrete problem for any specified parameter choice. Observing of course, that this does not translate into the solvability for the upstream general problem. The last Assumption (4) gives us the ability to later use an Autoencoder for a reduction.

Assumption 1. Let $\mathcal{S} := \{u_h^{\mu,\nu,t} \in \mathbb{R}^{N_h} \mid (\mu,\nu,t) \in \Theta \times \Theta' \times [0,T]\}$ denote the *total solution manifold* and $\mathcal{S}_{\mu,t} := \{u_h^{\mu,\nu,t} \in \mathbb{R}^{N_h} \mid \nu \in \Theta'\}$ denote the *solution submanifold* for a fixed set $(\mu,t) \in \Theta \times [0,T]$, then we assume that

1. the linear KnW of $\mathcal{S}$ decays slowly, i.e.

$$d_N(\mathcal{S}) \leq CN^{-\alpha},$$

where $0 < \alpha \leq 1$ and $C > 0$ some constant.

2. the linear KnW of $\mathcal{S}_{\mu,t}$ decays quickly for all $\mu \in \Theta$ and $t \in [0,T]$, i.e.

$$\sup_{(\mu,t)\in\Theta\times[0,T]} d_N(\mathcal{S}_{\mu,t}) \leq C' N^{-\beta},$$

where $\beta > 1$ and $C' > 0$ some constant.

This is mostly satisfied for problems in which the transport dominated or non-linear part is chained to the parameter μ, and any coersive or elliptic operator is most heavily influence by ν.

Assumption 2. Let Θ, Θ' and $[0,T]$ for $T > 0$ as above. We assume that all $N_s \in \mathbb{N}_{\geq 2}$ parameter data points according to Definition (3.1) are sampled uniformly and iid in the joint parameter space $\Theta \times \Theta'$, while a uniform grid is employed for the time variable, $\mathcal{T} = \{\Delta t, 2\Delta t, \ldots, N_t \Delta t\}$, where $N_t \in \mathbb{N}_{\geq 2}$, N_{s_1} and N_{s_2} are the respective number of random samples in the parameter spaces Θ and Θ', $N_s = N_{s_1} N_{s_2} \in \mathbb{N}$ is the total sample size, and $\Delta t = T/N_t$.

We remark, that the equidistant spacing of the time samples is a matter of convenience, and could be loosened under some constraints.

Assumption 3. Let $\mathcal{G} : \Theta \times \Theta' \times [0,T] \to \mathbb{R}^{N_h}$ be the parameter-to-solution map, mapping $(\mu,\nu,t) \mapsto u_h^{\mu,\nu,t}$. Here we assume that

1. $m = \operatorname*{ess\,inf}_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|u_h(\mu,\nu,t)\| > 0$, $M = \operatorname*{ess\,sup}_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|u_h(\mu,\nu,t)\| < \infty$,

2. $\mathcal{G}$ is Lipschitz-continuous with the constant $L > 0$.

A very important fact follows already directly from the presumably allowed well-posed definition of such a map; the solution $u_h(\cdot, \cdot; \mu, \nu)$ exists and is unique, in at least an L^2-embedded sense, since the time-derivate can be relaxed using a variational argument.

Assumption 4. Let $N \in \mathbb{N}$ be fixed. We assume that for s, s' sufficiently large, there are two maps $\psi_* : \mathbb{R}^n \to \mathbb{R}^N$ and $\psi'_* : \mathbb{R}^N \to \mathbb{R}^n$ which attain the perfect nonlinear embedding of the reduced solution manifold, as given in Definition (2.14) for any $n \leq 2(p+q)+3$, i.e. for $\mathcal{S}_N := \{q(\mu,\nu,t) = \mathbb{A}^T u(\mu,\nu,t) \,|\, (\mu,\nu,t) \in \Theta \times \Theta' \times \mathcal{T}\}$, where $\mathbb{A} \in \mathbb{R}^{N_h \times N}$ is the POD matrix,

$$\delta_n(\mathcal{S}_N) = 0.$$

Every following Reduced Order Model will pursue the universal (and vague) following goal.

Goal 1. Find

$$\hat{\mathcal{G}} : \Theta \times \Theta' \times [0,T] \to \mathbb{R}^{N_h}; \quad (\mu,\nu,t) \mapsto \hat{u}_h^{\mu,\nu,t},$$

such that for all $\mu \in \Theta$ and $\nu \in \Theta'$

$$\sup_{t\in[0,1]} \|u_h^{\mu,\nu,t} - \hat{u}_h^{\mu,\nu,t}\| = \sup_{t\in[0,1]} \|u_h(\cdot,t;\mu,\nu) - \hat{u}_h(\cdot,t;\mu,\nu)\|_{\mathcal{H}_h}$$

is small.

In general, we remind ourselves of the starting situation: $\mathcal{H}_h \subset \mathcal{H}$ is a finite dimensional approximation space of our weak problem. As mentioned before, we also have employed discretized time-steps. We therefore have something along the lines of

$$\begin{cases} M(\mu)\hat{\partial}_t u_h^{\mu,\nu}(t) + A(\mu,\nu)u_h^{\mu,\nu}(t) + N(u_h^{\mu,\nu}(t),\mu,\nu) = f(t;\mu,\nu) & t \in \mathcal{T}, \\ u_h^{\mu,\nu}(0) = u_0(\mu,\nu) & t = 0, \end{cases} \tag{3.4}$$

for $\hat{\partial}_t$ being a discrete approximation of the time-derivative and $u_h^{\mu,\nu}(t) \in \mathbb{R}^{N_h}$ for $t \in [t_k, t_{k+1})$ and some $k = 0, \ldots, N_t - 1$ the piecewise constant coefficient vector for the true solution

$$u_h(\cdot,t;\mu,\nu) = \sum_{j=1}^{N_h} [u_h^{\mu,\nu}(t)]_j \, \varphi_j(\mu) \in \mathcal{H}_h, \text{ for each } t \in [t_k, t_{k+1}) \text{ and } (\mu,\nu) \in \Theta \times \Theta'.$$

3.1 Intrusive Reduced Order Models

In this section we want to introduce some such *intrusive* methods for the dimensionality reduction of the finite dimensional parametric PDE. The name "intrusive" means to symbolize the "intrusion" of the method into the formulation of the PDE and the specifications of its operators. Coinciding with that, there are methods for linear and non-linear problems respectively, as the more simple and light weight reduction techniques can leverage an abstract linear variational problem to achieve a (Petrov-)Galerkin projection, whereas any non-linear problem, so a problem with $N \neq 0$ according to Equation (3.2), utilize the specifications of the non-linear operator to achieve reduction.

Remark 3.2 (Historic Literature)**.** The development of intrusive reduced order models (ROMs) dates back to the late 1970s and 1980s, when methods for projection onto low-dimensional trial spaces were systematically introduced in the context of parametrized PDEs. Early formulations of the *reduced basis (RB) method* can be found in the control and approximation theory literature, e.g. in the pioneering works of Courant and Hilbert on variational methods (Courant & Hilbert, 1953), and in the first computational implementations by Noor and coworkers in structural mechanics (Noor, 1980). These ideas were later rigorously developed into the modern RB methodology by Patera in the mid-1980s (Patera, 1984), and subsequently advanced by Maday, Rovas, Prud'homme and others in the 1990s and 2000s (Maday et al., 2002; Prud'homme et al., 2002).

In parallel, the POD emerged as a computational tool for obtaining optimal low-rank approximations of dynamical systems, building on statistical formulations dating back to Karhunen and Loève. The introduction of POD-Galerkin for intrusive model reduction was consolidated in applied mechanics and fluid dynamics in the 1990s and early 2000s (see (Holmes et al., 1996; Kunisch & Volkwein, 2001)).

As nonlinear problems became central, the cost of evaluating nonlinear terms prompted the development of *hyper-reduction techniques*, such as the Empirical Interpolation Method (EIM) (Barrault et al., 2004) and its discrete variants (DEIM) (Chaturantabut & Sorensen, 2010), which remain core components of modern intrusive ROM frameworks.

We will start with the framework of projection-based methods, that necessitate a problem with $N = 0$.

All of the following methods will be discussed broadly, as the thesis concerns itself mostly with the later data-driven ROMs.

Definition 3.3 (Reduced Basis)**.** Let $\mathcal{H}_n \subset \operatorname{span}(u_h(\cdot, t; \mu, \nu) \in \mathcal{H}_h \,|\, (\mu, \nu, t) \in \mathcal{P}_1 \times \mathcal{P}_2 \times \mathcal{T})$ be an n-dimensional subspace of $\mathcal{H}_h$, then if $\{\hat{\varphi}_j\}_{j=1}^n$ is an orthonormalized basis of $\mathcal{H}_n$, we call this the *reduced basis.*

From this starting point, the goal of all *reduced basis methods* is to find a reduced basis $\{\hat{\varphi}_j(\mu)\}_{j=1}^n$ and $\hat{u}_h^{\mu,\nu}(t) \in \mathbb{R}^n$ for $n \ll N_h$, such that

$$\hat{u}_h(\cdot, t; \mu, \nu) = \sum_{j=1}^{n} [\hat{u}_h^{\mu,\nu}(t)]_j \, \hat{\varphi}_j(\mu) \in \mathcal{H}_n \subset \mathcal{H}_h,$$

where, in a yet vague formulation,

$$\hat{u}_h(\cdot, t; \mu, \nu) \approx u_h(\cdot, t; \mu, \nu) \quad \text{for all } \mu \in \Theta, \nu \in \Theta' \text{ and } t \in [0, T].$$

In the terminology the RB methods are a setup for many different ROMs, but most often appear for so-called projection-based ROMs, which itself are intrusive ROMs. They help with the identification of a suitable basis to project upon. Two prominent examples of such reduced basis methods in the context of projection-based ROMs can be found in the following. Both will enhibit intrinsically an *offline* and an *online* part in the algorithm, whose separation we will mostly neglect. However, when remarking something about the offline procedure, we usually talk about the part of the algorithm that can be collected a priori to its purposed deployment. Here this is the collection and reduction of dimension using a reduced basis via samples. On the other hand, online refers to its counterpart; the many-query element of the procedure, such as the evaluation of the best-guess parameter-to-solution map. This terminology is used often throughout the literature as it instinctively makes sense in a software development environment.

3.1.1 Galerkin Proper Orthogonal Decomposition

As a prerequisit for Galerkin orthogonality, we assume here that the weak form giving rise to the equation at hand derives from an abstract bilinear form in the stationary part. This is justified if the non-linear summand is zero. The source for this subsection is (Quarteroni et al., 2015, p. 137).

The POD as introduced in Definition (2.10) is one of the most common methods for a reduction in a classical sense. We can use the system above to solve for the solution using a Galerkin projection for the following Ansatz:

$$\hat{u}_h(x,t;\mu,\nu) = \sum_{j=1}^{n} [\hat{u}_h^{\mu,\nu}(t)]_j \, \xi_j(x), \tag{3.1}$$

where $x \in \Omega$, $t \in [0,1]$, $(\mu,\nu) \in \Theta \times \Theta'$ and ξ_j for $1 \leq j \leq n$ are the POD modes, as given by the construction of the POD in Defintition (2.10) with $\mathbb{G} := \mathbb{I}_{N_h}$, and $\hat{u}_h^{\mu,\nu}(t) \in \mathbb{R}^n$ is the coefficient vector, that solves the following reduced equation

$$\begin{cases} \mathbb{A}^T M(\mu)\mathbb{A}\hat{\partial}_t \hat{u}_h^{\mu,\nu}(t) + \mathbb{A}^T A(\mu,\nu)\mathbb{A}\hat{u}_h^{\mu,\nu}(t) = \mathbb{A}^T f(t;\mu,\nu) & t \in \mathcal{T}, \\ \hat{u}_h^{\mu,\nu}(0) = u_0(\mu,\nu) & t = 0. \end{cases} \tag{3.2}$$

Note, that in this construction $\mathbb{A}$ is *not* $M(\mu)$-orthogonal, thus losing certain projection qualities with regard to the inherent L^2-norm of the underlying Hilbert space $\mathcal{H}_h$. The parameter-to-solution map can be recovered by the following Algorithm (2).

Algorithm 2 POD-Galerkin

1: **Input:** $(\mu_j,\nu_j)_{j=1}^{N_s}$ data sample and $n < N_h$ reduced dimension
2: **Output:** approximated parameter-to-solution-map $(\mu,\nu) \mapsto \hat{u}_h(\cdot,\cdot;\mu,\nu)$
3: Compute $\{u_h^{\mu_j,\nu_j,t_{k_m}}\}_{m=1}^{N_t}$ from Equation (3.4) (with $N = 0$)
4: $(\xi_j,\sigma_j)_{j=1}^{n} \leftarrow \mathrm{POD}(u_h^{\mu_j,\nu_j,t_{k_m}}, n)$
5: $\mathbb{A} \leftarrow \mathrm{stack}(\xi_j, j = 1,\ldots,n)$
6: Build reduced Equation (3.2)
7: For each (μ,ν) solve Equation (3.2)
8: Recover $\hat{u}_h(\cdot,\cdot;\mu,\nu)$ via Equation (3.1)

3.1.2 Greedy Proper Orthogonal Decomposition

An introduction to this method in the stationary and well-posed case can be found in (Siena et al., 2022). As done previously in the subsection above, we assume $N = 0$ in Equation (3.4). This gives us a coersive abstract bilinear form for the stationary part. Additionally let $\eta : \Theta \times \Theta' \to \mathbb{R}$ be a function estimating the error of projection under a specific parameter set, i.e. such that

$$\sup_{t\in\mathcal{T}} \|u_h(\mu,\nu,t) - \hat{u}_h(\mu,\nu,t)\| \leq \eta(\mu,\nu),$$

for $(\mu,\nu) \in \mathcal{P}$. Purely using POD-Galerkin can encounter major pitfalls, namely if the range of $\mathcal{P}$ is extensive, since then calculating a singluar value decomposition for the correlation matrix might get too cumbersome to provide all areas of interest in the set. Moreover, to obtain the snapshots for fixed parameters, a complete time-trajectory must be computed, which entails a lot of overhead of otherwise useless information. To overcome this (Grepl & Patera, 2005) proposed, and later (Haasdonk & Ohlberger, 2008) improved upon, the mixed POD-Greedy approach, which takes a POD over the time-trajectories (or, in the improved version, the error trajectories) while using the Greedy approach to handle the parameter samples. It follows a similar workflow, as the POD-Galerkin, since the only thing, that changes is the choice of basis, and the fact that the size of the basis is chosen according to the tolerance in the offline part of the algorithm, i.e.

$$\hat{u}_h(x,t;\mu,\nu) = \sum_{j=1}^{N} [\hat{u}_h^{\mu,\nu}(t)]_j \, \xi_j(x), \tag{3.3}$$

with $x \in \Omega$, $t \in [0,1]$, $(\mu, \nu) \in \Theta \times \Theta'$ identifies the reduced solution of dimension N, where ξ_j are the basis functions obtained from the offline procedure in Algorithm (3) for $1 \leq j \leq N$ and the coefficient vector being the solution to

$$\begin{cases} \mathbb{V}_{\mathrm{rb}}^T M(\mu)\mathbb{V}_{\mathrm{rb}}\hat{\partial}_t \hat{u}_h^{\mu,\nu}(t) + \mathbb{V}_{\mathrm{rb}}^T A(\mu,\nu)\mathbb{V}_{\mathrm{rb}}\hat{u}_h^{\mu,\nu}(t) = \mathbb{V}_{\mathrm{rb}}^T f(t;\mu,\nu) & t \in \mathcal{T}, \\ \hat{u}_h^{\mu,\nu}(0) = u_0(\mu,\nu) & t = 0. \end{cases} \tag{3.4}$$

Note again, that $\mathbb{V}_{\mathrm{rb}}$ is *not* $M(\mu)$-orthonormal. The complexity of the offline computation for this Algorithm (3) is additive, and not multplicative, in the number of training data $\mathcal{P}$, which is a huge advantage when compared to the direct POD-Galerkin approach.

Algorithm 3 POD-Greedy-Galerkin

Input: `tol`, $\mathcal{Z} = 0$, $N = 0$, $(\mu_j, \nu_j)_{j=1}^{N_s}$, N_1 and N_2
Output: approximated parameter-to-solution-map $(\mu,\nu) \mapsto \hat{u}_h(\cdot,\cdot;\mu,\nu)$
for $j = 1, \dots, N_s$ **do**
  Compute $\{u_h^{\mu_j,\nu_j,t_{k_m}}\}_{m=1}^{N_t}$ from Equation (3.4) (with $N = 0$)
  $(\zeta_j, \lambda_j)_{j=1}^{N_1} \leftarrow \mathrm{POD}(\{u_h^{\mu_j,\nu_j,t_{k_m}}\}_{m=1}^{N_t}, N_1)$
  $\mathcal{Z} \leftarrow \{\mathcal{Z}, \{\zeta_1, \dots, \zeta_{N_1}\}\}$
  $N \leftarrow N + N_2$
  Compress: $(\xi_j, \sigma_j)_{j=1}^{n} \leftarrow \mathrm{POD}(\mathcal{Z}, N)$
  $(\mu_{j+1}, \nu_{j+1}) \leftarrow \underset{(\mu,\nu)\in\mathcal{P}}{\arg\max}\, \eta(\mu,\nu)$
  if $\eta(\mu_{j+1}, \nu_{j+1}) >$ `tol` **then**
    $j \leftarrow j + 1$
  else
    break
  end if
end for
$\mathbb{V}_{\mathrm{rb}} \leftarrow \mathrm{stack}(\xi_1, \dots, \xi_N)$
Build reduced Equation (3.4)
For each (μ,ν) solve Equation (3.4)
Recover $\hat{u}_h(\cdot,\cdot;\mu,\nu)$ via Equation (3.3)

Remark 3.4. It would be possible to include the physical parameter into the first POD scheme in each iteration, which could give a high yield for this specific problem as the Kolmogorov n-width decay as imposed by Assumption (1) implies. However since we lose generality by setting $N = 0$ in both projection-based algorithm, this is only an argument for divergent discussions.

3.1.3 Non-linear Reduced Order Models

Add some literature

3.2 Data-driven Reduced Order Models

In contrast to the introduced intrusive ROMs above, the *data-driven* alternatives aim to circumvent the direct use of the governing equations, and, as such, allow for more leniency toward unknown operator qualities.

Remark 3.5 (Historic Literature)**.** The origins of data-driven ROMs can be traced back to techniques in fluid dynamics and control, where the objective was to extract coherent structures or dominant modes from experimental measurements without explicit reliance on the full Navier–Stokes equations. A first influential step was the development of the POD for experimental data, introduced in the turbulence community by Lumley (Lumley, 1967), and further popularized in the coherent structure analysis of Sirovich (Sirovich, 1987). While POD itself does not provide a dynamical law for the online phase, it laid the foundation for representing complex systems with a small number of empirical modes.

A decisive breakthrough came with the advent of the *Dynamic Mode Decomposition* (DMD) (Schmid, 2010), which may be interpreted as a data-driven approximation of the infinite-dimensional Koopman operator (Rowley et al., 2009). DMD represents temporal evolution as a best-fit linear operator acting on snapshot data, thereby yielding both modal decompositions and a surrogate model for prediction. This was one of the first systematic attempts to construct a fully non-intrusive ROM.

In parallel, regression-based approaches have been developed to learn reduced operators directly from data, often under the name of *operator inference* (Peherstorfer et al., 2016), or in the form of interpolation-based reduced models, e.g. POD with interpolation (PODI) as in Barrault et al. (Barrault et al., 2004). These methods demonstrate that the reduced dynamics can be inferred without Galerkin projection, either by statistical regression or by interpolation in parameter space.

For this section we intentionally do not give an explicit training algorithm for any of the methods presented, since it can be infered with an optimizer of the readers choice from the discussion in Subsection (2.3.5). With readability in mind, we thus only state the architecture of the network at hand and its loss function. Most importantly, we assume that some choice of basis for $\mathcal{H}_h$ is already done a priori to the training of the network and independent of $\mu \in \Theta$ thus setting $\mathbb{G} := M(\mu)$ for all $\mu \in \Theta$ as our mass matrix. Under Equation (3.3) this also implies a μ-independent norm $\|\cdot\|$, which in fact allow us to use the POD matrix, if constructed with $\mathbb{G}$, as a true projector onto the span of $\mathbb{A}$ in an L^2 sense.

For clarity, we formulate the following as an assumption, since from experience, the reduction system across comparable architectures can get confusing, and the reader is therefore encouraged to revisit this assumption, whenever the general setting appears to be unclear.

Assumption 5 (Dimension Hierarchy)**.** Let for all following data-driven Reduced Order Models, the following natural number hierarchy be fixed

$$n < N < N_A \ll N_h.$$

Note, that "$\ll$" is an unspecified relation, meaning to symbolize a "large" difference.

A general Deep Learning based ROM (DL-ROM) first appeared in architecture and training in the paper (Fresca et al., 2020); to properly approximate the potentially non-linear solution manifold of the high fidelity FOM, we employ an Autoencoder (AE), and on the reduced manifold we employ some other Deep Learning architecture, like the proposed Deep Feed-Forward Neural Network (DFNN), to learn the latent dynamics of the parametric PDE. This concludes in the following approximation of the FOM coefficient solution $u_h^{\mu,\nu,t} \in \mathbb{R}^{N_h}$ for each $\mu \in \Theta$, $\nu \in \Theta'$ and $t \in \mathcal{T}$

$$\hat{u}_h^{\mu,\nu,t} = \left(R_\rho(\Psi_h(\theta^*_{\mathrm{D}})) \circ R_\rho(\Phi_n(\theta^*_{\mathrm{DF}}))\right)(t;\mu,\nu) \in \mathbb{R}^{N_h}, \tag{3.1}$$

where we set $\Psi_h = f_h^{\mathrm{D}}$ for the decoder architecture with input dimension n and output dimension N_h (and $\Psi'_n = f_n^{\mathrm{E}}$ the corresponding encoder architecture) of a convolutional AE. Then Ψ'_n denotes the projection onto the reduced trial solution manifold and we let $\Phi_n = \Phi_n^{\mathrm{DF}}$ to be the DFNN architecture of input dimension $p+q+1$ and output dimension n, mapping the dynamics of the parametric PDE onto the reduced trial solution manifold.

The subsequent training is then performed via a loss function of a convex combination of both, the projection error using the encoder and the dynamics error considering both the decoder and the feed forward network, i.e. for $\theta = (\theta_D, \theta_E, \theta_{DF})$ define the optimal weights via the problem in Definition (2.34) with the loss function

$$\begin{aligned}
&\mathcal{L}_{\text{DL-ROM}}\left(R_\rho(\Psi_h \odot \Phi_n(\theta))|\{\mu_i,\nu_j,k\Delta t\}_{i,j,k},\{u_h^{\mu_i,\nu_j,k\Delta t}\}_{i,j,k}\right) = \\
&\qquad\sum_{i=1}^{N_{s_1}}\sum_{j=1}^{N_{s_2}}\sum_{k=1}^{N_t}\frac{\omega_h}{2}\|u_h^{\mu_i,\nu_j,k\Delta t} - \left(R_\rho(\Psi_h(\theta_{\mathrm{D}}))\circ R_\rho(\Phi_n(\theta_{\mathrm{DF}}))\right)(\mu_i,\nu_j,k\Delta t)\|^2 \\
&\qquad\qquad + \frac{1-\omega_h}{2}|R_\rho(\Psi_n'(\theta_{\mathrm{E}}))(u_h^{\mu_i,\nu_j,k\Delta t}) - R_\rho(\Phi_n^{\mathrm{DF}}(\theta_{\mathrm{DF}}))(\mu_i,\nu_j,k\Delta t)|^2.
\end{aligned} \tag{3.2}$$

3.2.1 Proper Orthogonal Decomposition Deep Learning ROM

The main idea of the POD-DL-ROM is the additional employment of an outer a priori reduction of DL-ROM problem using a POD. It can be found in (Fresca & Manzoni, 2022). Note, that we sway a bit from the source, since we are trying to achieve an estimate, which is close to the high-fidenlity solution in an L^2-sense and not a euklidian one by applying the mass matrix from the start of the section. Such a reduction yields great benefits, such as the scalability in N_h and reduction in computational cost for the offline training. It is done by employing a POD (see Algorithm (1)) on the stacked solution trajectories

$$U := \left[u_h^{\mu_1,\nu_1}|\cdots|u_h^{\mu_{N_{s_1}},\nu_1}|\cdots|u_h^{\mu_{N_{s_1}},\nu_{N_{s_2}}}\right]^T \in \mathbb{R}^{N_h\times N_{\text{data}}},$$

with the solution trajectories of Definition (3.1), resulting in the POD matrix $\mathbb{A}\in\mathbb{R}^{N_h\times N}$ for the $N < N_h$ from Assumption (5). Then we can define the pre-reduced solution snapshots by computing

$$u_{\text{red}}^{\mu,\nu,t} := \mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\in\mathbb{R}^N \qquad \text{for all } (\mu,\nu,t)\in\mathcal{P}_1\times\mathcal{P}_2\times\mathcal{T}.$$

By $\mathbb{G}$-orthonormality of $\mathbb{A}$, we can thus define the FOM approximate coefficient solution $\hat{u}_h^{\mu,\nu,t}\in\mathbb{R}^{N_h}$ for each $\mu\in\Theta$, $\nu\in\Theta'$ and $t\in\mathcal{T}$

$$\hat{u}_h^{\mu,\nu,t} = \mathbb{A}\cdot\left(R_\rho(\Psi_A(\theta_{\mathrm{D}}^*))\circ R_\rho(\Phi_n(\theta_{\mathrm{DF}}^*))\right)(t;\mu,\nu)\in\mathbb{R}^{N_h}, \tag{3.3}$$

where we set $\Psi_N = f_N^{\mathrm{D}}$ for the decoder architecture with input dimension n and output dimension N (and $\Psi_n' = f_n^{\mathrm{E}}$ the corresponding encoder architecture) of a convolutional AE. Then f_n^{E} denotes the projection onto the reduced trial solution manifold and we let $\Phi_n = \Phi_n^{\mathrm{DF}}$ to be the DFNN architecture of input dimension $p+q+1$ and output dimension n same as before.

The loss function has now the attribute of being fed only at most N dimensional training data, and only evaluating at most N dimensional euklidian norms, hence greatly increasing performance, i.e. with P being the network parallelization operator

$$\begin{aligned}
&\mathcal{L}_{\text{POD-DL-ROM}}\left(R_\rho(P(\Psi_N\odot\Phi_n,\Psi_n')(\theta))|\{\mu_i,\nu_j,k\Delta t\}_{i,j,k},\{u_{\text{red}}^{\mu_i,\nu_j,k\Delta t}\}_{i,j,k}\right) = \\
&\qquad\frac{1}{N_{\text{data}}}\sum_{i=1}^{N_{s_1}}\sum_{j=1}^{N_{s_2}}\sum_{k=1}^{N_t}\left(\frac{\omega_h}{2}|u_{\text{red}}^{\mu_i,\nu_j,k\Delta t} - \left(R_\rho(\Psi_N(\theta_{\mathrm{D}}))\circ R_\rho(\Phi_n(\theta_{\mathrm{DF}}))\right)(\mu_i,\nu_j,k\Delta t)|^2\right. \\
&\qquad\qquad\left. + \frac{1-\omega_h}{2}|R_\rho(\Psi_n'(\theta_{\mathrm{E}}))(u_{\text{red}}^{\mu_i,\nu_j,k\Delta t}) - R_\rho(\Phi_n(\theta_{\mathrm{DF}}))(\mu_i,\nu_j,k\Delta t)|^2\right).
\end{aligned} \tag{3.4}$$

add a justification of the use of the reduced space and euklidian norm as done in franco2024

3.2.2 Deep Orthogonal Decomposition based ROMs

A big advantage of the POD-DL-ROM is the fact, that under the assumption of a static basis $\{\xi_j\}_{j=1}^{N_h} \subset \mathcal{H}_h$ of $\mathcal{H}_h$ and its associated mass matrix $\mathbb{G}$ and $\mathcal{H}_h$ equivalent norm $\|\cdot\|$, we may reduce the high dimensional loss function in its essence to a managable computational complexity. However, this so far ignored the particular problem in sight; it ignores Assumption (1), which guarantees a significant reduction cost, when employing a (μ, t)-independent projector. To circumvent this Kolmogorov N-width barrier for $N \ll N_h$, one can instead of using a POD for the choice of reduced basis, invoke a Neural Network based linear projector, that fits a basis superposition dependend on μ and t.

The following technically depicts a RB method.

Definition 3.6 (Deep Orthogonal Decomposition)**.** Collect a POD (according to Algorithm (1)) with $\mathbb{G}$ given as above using the stacked solution trajectories

$$U := \left[u_h^{\mu_1,\nu_1}\middle|\cdots\middle|u_h^{\mu_{N_{s1}},\nu_1}\middle|\cdots\middle|u_h^{\mu_{N_{s1}},\nu_{N_{s2}}}\right]^T \in \mathbb{R}^{N_h \times N_{\text{data}}},$$

with the solution trajectories of Definition (3.1), resulting in the POD matrix $\mathbb{A} \in \mathbb{R}^{N_h \times N_A}$ for some $N_A \leq N_h$. Now set up for the inner module of the Deep Orthogonal Decomposition a Neural Network architecture $\Phi_{\widetilde{\mathbb{V}}}$ with input dimension $p+1$ and output dimension $N_A \times N$ for some $N < N_A$ such that

$$\Phi_{\widetilde{\mathbb{V}}} = \text{ORTH} \circ \text{STACK} \circ (P((R_1 \odot s), \ldots, (R_N \odot s)))(\mu, t),$$

where P is denoting the network parallelization operator as given in Definition (2.24)

1. s is a Deep Feed Forward Neural Network architecture with input dimension $p+1$ and output dimension l,
2. $R_1, \ldots, R_N$ are a collection of Deep Feed Forward Neural Network architectures with input dimensions l and output dimensions N_A,
3. $\text{STACK} : \mathbb{R}^{N_A \cdot N} \to \mathbb{R}^{N_A \times N}$ is a canonical stacking operator,
4. $\text{ORTH} : \mathbb{R}^{N_A \times N} \to \mathbb{R}^{N_A \times N}$ is an orthonormalization unit, that orthonormalizes any given matrix $W \in \mathbb{R}^{N_A \times N}$ into $\widetilde{W} \in \mathbb{R}^{N_A \times N}$ such that $\text{span}(W) = \text{span}(\widetilde{W})$. This is most often done using a QR decomposition (Golub & Van Loan, 2013).

The realization of the network $\Phi_{\widetilde{\mathbb{V}}}$ for some $\mu \in \Theta$, $t \in [0, T]$ and any chosen neural network weight $\theta_{\text{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ according to Definition (2.34) will be abbreviated with

$$\widetilde{\mathbb{V}}_{\mu,t}(\theta_{\text{DOD}}) := R_\rho(\Phi_{\widetilde{\mathbb{V}}}(\theta_{\text{DOD}}))(\mu, t). \tag{3.5}$$

For any choice of weights, this is called the *inner module of the Deep Orthogonal Decomposition*, or in short the inner DOD module.

The Deep Orthogonal Decomposition can hence be achieved by upward projection under the POD matrix $\mathbb{A}$ of the inner DOD module, i.e. a realization of the underlying DOD neural network for a choice of weights $\theta_{\text{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ is attained for some $\mu \in \Theta$ and $t \in [0, T]$ by

$$\mathbb{V}_{\mu,t}(\theta_{\text{DOD}}) := \mathbb{A} \cdot \widetilde{\mathbb{V}}_{\mu,t}(\theta_{\text{DOD}}) \in \mathbb{R}^{N_h \times N}. \tag{3.6}$$

Nonetheless this does not warrant a "good" DOD, just a defined one. For this one needs to assert a qualitative loss function and some data, which we will get by pre-reduction in the same manner as before, even though in contrast $N_A > N$ in general, i.e.

$$u_{\text{pre-red}}^{\mu,\nu,t} := \mathbb{A}^T \mathbb{G} u_h^{\mu,\nu,t} \in \mathbb{R}^{N_A} \qquad \text{for all } (\mu, \nu, t) \in \mathcal{P}_1 \times \mathcal{P}_2 \times \mathcal{T}.$$

Then the loss function with regard to that data can be defined as follows

$$\begin{aligned} &\mathcal{L}_{\text{DOD}} \left(R_\rho(\Phi_{\widetilde{\mathbb{V}}}(\theta_{\text{DOD}})) | \{\mu_i, \nu_j, k\Delta t\}_{i,k}, \{u_{\text{pre-red}}^{\mu_i,\nu_j,k\Delta t}\}_{i,j,k} \right) \\ &\qquad = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} |u_{\text{pre-red}}^{\mu_i,\nu_j,k\Delta t} - \widetilde{\mathbb{V}}_{\mu_i,k\Delta t}(\theta_{\text{DOD}}) \widetilde{\mathbb{V}}_{\mu_i,k\Delta t}^T(\theta_{\text{DOD}}) u_{\text{pre-red}}^{\mu_i,\nu_j,k\Delta t}|^2. \end{aligned} \tag{3.7}$$

Note, that this is indeed again a N_A-dimensional loss function, thanks to the $\mathbb{G}$-orthonormality of the POD matrix. In some abuse of notation, we will sometimes refer to the optimized version of the DOD network as the DOD.

Notice, that we have not yet performed any actual search for the approximate parameter-to-solution map, that maps a specific instance of parameters to a coefficient time-trajectory, that can then be used to build the solution map in the underlying high fidelity Hilber space approximating the FOM solution in Equation (3.1).

In this paper, we propose two novel data-driven ROMs, based on the DOD, to find an approximate solution-to-parameter map. These are the *Deep Orthogonal Decomposition with Deep Neural Network* (DOD+DNN) and the *Deep Orthogonal Decomposition based Deep Learning ROM* (DOD-DL-ROM). The former indeed symbolizes a general approach hence including the latter, but, with some abuse of notation, we admit the DOD+DNN a naive direct Deep Feed Forward architecture for the determination of the coefficient, while the DOD-DL-ROM is an adaptation of the DL-ROM architecture introduced in the header of the section.

Let in general for the following $n < N \ll N_A \leq N_h$ be natural numbers.

DOD+DNN

Let $\Phi_N := \Phi_N^{\text{DF}}$ be some Deep Feed-Forward Neural Network architecture with input dimension $p + q + 1$ and output dimension N, then the approximate parameter-to-solution map for a fixed weight $\theta = (\theta_{\text{DOD}}, \theta_{\text{DF}}) \in \Theta(\Phi_{\widetilde{\mathbb{V}}}) \times \Theta(\Phi_n^{\text{DF}})$ according to Definition (2.34) can be given for each $(\mu, \nu, t) \in \Theta \times \Theta' \times [0, T]$ via

$$\hat{u}_h^{\mu,\nu,t} = \mathbb{V}_{\mu,t}(\theta_{\text{DOD}}) \cdot R_\rho(\Phi_N(\theta_{\text{DF}}))(\mu, \nu, t) \in \mathbb{R}^{N_h}. \tag{3.8}$$

Subsequently the training is done in series and not in parallel, as the optimal DOD parameter θ_{DOD}^* needs to be found first. Secondly, the $\mathbb{G}$-orthonormality of the DOD can be leveraged with

$$u_{\text{red}^2}^{\mu,\nu,t} := \mathbb{V}_{\mu,t}(\theta_{\text{DOD}}^*)^T \mathbb{G} u_h^{\mu,\nu,t} \in \mathbb{R}^N \qquad \text{for all } (\mu, \nu, t) \in \mathcal{P}_1 \times \mathcal{P}_2 \times \mathcal{T},$$

to further reduce the training data and finally define the loss function for the DOD+DNN architecture

$$\begin{aligned} &\mathcal{L}_{\text{DOD+DNN}} \left(R_\rho(\Phi_N(\theta_{\text{DF}})) | \{\mu_i, \nu_j, k\Delta t\}_{i,k}, \{u_{\text{red}^2}^{\mu_i,\nu_j,k\Delta t}\}_{i,j,k} \right) \\ &\qquad = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} |u_{\text{red}^2}^{\mu_i,\nu_j,k\Delta t} - R_\rho(\Phi_N(\theta_{\text{DF}}))(\mu_i, \nu_j, k\Delta t)|^2. \end{aligned} \tag{3.9}$$

This is a rather obvious extension of (Franco et al., 2024), but works surprisingly well. It will be the basic benchmark to overcome for the following algorithms.

DOD-DL-ROM

Let $\Psi_N := f_N^{\mathrm{D}}$ be a decoder architecture with input dimension n and output dimension N (and $\Psi'_n := f_n^{\mathrm{E}}$ the corresponding encoder architecture), and $\Phi_n := \Phi_n^{\mathrm{DF}}$ be some Deep Feed-Forward Neural Network architecture with input dimension $p + q + 1$ and output dimension $n$then the approximate parameter-to-solution map for a fixed weight $\theta = (\theta_{\mathrm{DOD}}, \theta_{\mathrm{D}}, \theta_{\mathrm{DF}}) \in \Theta(\Phi_{\widetilde{\mathbb{V}}}) \times \Theta(f_N^{\mathrm{D}}) \times \Theta(\Phi_n^{\mathrm{DF}})$ according to Definition (2.34) can be given for each $(\mu, \nu, t) \in \Theta \times \Theta' \times [0, T]$ via

$$\hat{u}_h^{\mu,\nu,t} = \mathbb{V}_{\mu,t}(\theta_{\mathrm{DOD}}) \cdot (R_\rho(\Psi_N(\theta_{\mathrm{D}})) \circ R_\rho(\Phi_n(\theta_{\mathrm{DF}})))\,(t; \mu, \nu) \in \mathbb{R}^{N_h} \tag{3.10}$$

Same, as for the DOD+DNN, the training is done in series. One observes, that the training data can be again reduced with

$$u_{\mathrm{red}^2}^{\mu,\nu,t} := \mathbb{V}_{\mu,t}(\theta^*_{\mathrm{DOD}})^T \mathbb{G} u_h^{\mu,\nu,t} \in \mathbb{R}^N \qquad \text{for all } (\mu, \nu, t) \in \mathcal{P}_1 \times \mathcal{P}_2 \times \mathcal{T},$$

and then used to define the DOD-DL-ROM loss function utilizing the $\mathbb{G}$-orthonormality of the DOD, hence

$$\begin{aligned}\mathcal{L}_{\text{DOD-DL-ROM}}&\left(R_\rho(P(\Psi_N \odot \Phi_n, \Psi'_n)(\theta)) | \{\mu_i, \nu_j, k\Delta t\}_{i,j,k}, \{u_{\mathrm{red}^2}^{\mu_i,\nu_j,k\Delta t}\}_{i,j,k}\right) = \\ &\frac{1}{N_{\mathrm{data}}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} \left(\frac{\omega_h}{2} \left| u_{\mathrm{red}^2}^{\mu_i,\nu_j,k\Delta t} - (R_\rho(\Psi_N(\theta_{\mathrm{D}})) \circ R_\rho(\Phi_n(\theta_{\mathrm{DF}})))\,(\mu_i, \nu_j, k\Delta t) \right|^2 \right. \\ &\qquad \left. + \frac{1 - \omega_h}{2} \left| R_\rho(\Psi'_n(\theta_{\mathrm{E}}))(u_{\mathrm{red}^2}^{\mu_i,\nu_j,k\Delta t}) - R_\rho(\Phi_n^{\mathrm{DF}}(\theta_{\mathrm{DF}}))(\mu_i, \nu_j, k\Delta t) \right|^2 \right).\end{aligned} \tag{3.11}$$

Notice, that this reduction allows us to elect an algorithm, which uses only the n- and N-dimensional euklidian norm, therefore enabling even less computational complexity for each training epoch, under the assumption of an already optimized DOD.

3.2.3 CoLoRA-DL-ROM

Introduce CoLoRA DL-ROM

Here we aim at minimizing the dimension through two major reductions; the first being the reduction of the stationary DOD to the latent dimension of the μ-invariant solution manifold of the stationary equivalent to the problem (regarding the Dirichlet boundary conditions as the initial conditions on some a priori defined boundary), and the second concerning the actual reduction with CoLoRA-layered architecture to the latent dynamic dimension $n << N_h$. The general workflow is thus given by

$$\hat{u}_h(t; \mu, \nu, \theta^*_{sDOD}, \theta_C) = \mathbb{A} \cdot \mathcal{C}_L(\ldots(\mathcal{C}_2(\mathcal{C}_1(V_0(\mu, \theta^*_{sDOD}) \cdot \phi_0(\mu, \nu, \theta^*_{sDOD})), \theta_{C_1}), \theta_{C_2}), \theta_{C_L}), \tag{3.12}$$

where $\mathcal{C}_i : \mathbb{R}^{N_A} \times \Theta_C \to \mathbb{R}^{N_A}; \mathcal{C}_i(x, \theta_{C_i}) = W_i x + A_i \operatorname{diag}(\alpha_i(\nu, t)) B_i x + b_i$ for all $1 \le i \le L$. Remark, that we will still employ a pre-reduction using the rPOD matrix $\mathbb{A}$ stemming from the same snapshot matrix S as in both Options before.

Chapter 4

Approximation Theory for Reduced Order Models

This chapter provides the main results of this thesis based on the data-driven ROMs introduced in Chapter (3). As such we repeat, that in order to execute the reduction for the norm $\|\cdot\|$ defined in that chapter, we require a mass matrix, that is independent of $\mu \in \Theta$, since otherwise the orthonormality with regard to it cannot be computed under the generalized POD Algorithm (1). However, we note, that the paper (Brivio et al., 2023) we base most of this chapter on, and severall other works dealing with a similar setup, instead neglect the representation in the underlying Hilbert space and use a simple euklidian norm for the coefficient vectors. For our sitution though, we let

$$\hat{u}_h(x, t; \mu, \nu) = \sum_{j=1}^{N_h} [\hat{u}_h^{\mu,\nu}(t)]_j \varphi_j(x), \tag{4.1}$$

for all $x \in \Omega$, $t \in [0, T]$ and parameters $(\mu, \nu) \in \Theta \times \Theta'$, which simultaneously specifies $[\mathbb{G}]_{ij} = \langle \varphi_i, \varphi_j \rangle = [M(\mu)]_{ij}$ as the mass matrix. The norm for this chapter will be, if not stated otherwise, the norm from Equation (3.3), so here

$$\|u\| := \sqrt{u^T \mathbb{G} u},$$

for all $u \in \mathbb{R}^{N_h}$. Furthermore the solution vector $\hat{u}_h^{\mu,\nu}(t) = \hat{u}_h^{\mu,\nu,t} \in \mathbb{R}^{N_h}$ is set to be constant between the time instants in our model, i.e. for all $t \in [(k-1)\Delta t, k\Delta t)$ for some $k = 1, \ldots, N_t$. As a goal of this chapter, we will provide upper bounds on the complexity of the networks making up the data-driven ROMs, where we especially outline a comparison between the POD-DL-ROM and the newly introduced DOD-DL-ROM.

In order to analyse any error occuring from a data-based approximation, we need to set up a probability space, in whose sense a convergence under an increased data set can be viewed. This is to be regarded as a clarification of Assumption (2).

Definition 4.1 (Sampling)**.** Let $(\mathcal{P}, \mathcal{B}(\mathcal{P}))$ be a Borel-measureable parameter space and $(\Omega_{\mathbb{P}}, \mathcal{F}, \mathbb{P})$ some underlying probability space. Then we refer to an *independent sampling of size* N, if for each $i \in \{1, \ldots, N\}$ the random variable mapping attaining a sample (μ_i, ν_i)

$$X_i : (\Omega_{\mathbb{P}}, \mathcal{F}, \mathbb{P}) \to (\mathcal{P}, \mathcal{B}(\mathcal{P}))$$

is measurable, its push forward $\mathbb{P}^{X_i}$ admits the same distribution $\mathbb{P}^X$, i.e.

$$\mathbb{P}[X_i \in A] = \mathbb{P}^{X_i}[A] = \mathbb{P}^X[A],$$

and the samples are independent. We further name it *iid sampling*, if the distribution $\mathbb{P}^X$ is the uniform distribution on $\mathcal{P}$.

With this in mind, we can view sampling as the realization of a random variable, that provides an element of the sample space. Therefore any mention of a probability space or a probability measure $\mathbb{P}$ refers to the samplying procedure in the probability space as given in Definition (4.1).

The upcoming proposition is a necessary result for the following chapters, as it provides us with a practical bound on the needed number of samples.

Proposition 4.2. Let $f \in L^2(\Theta \times \Theta' \times [0,T])$. Under Assumption (2), it is true, that

$$\mathbb{E}\left[\left| \int_{\Theta\times\Theta'\times[0,T]} f(\mu,\nu,t)d(\mu,\nu,t) - \frac{|\Theta\times\Theta'\times[0,T]|}{N_{data}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} f(\mu_i,\nu_j,k\Delta t) \right|\right] \leq \mathcal{O}(N_s^{-1/2}+N_t^{-1}).$$

prove this

4.1 Error Decomposition Formulas

To adequately start analysing the ROMs in question, one needs to recognize the different aspects of "error" inherent to their deployment. Namely, we can subdivide these seperate instances of inaccuracy occuring in the process of both, the POD-based data-driven ROM and the DOD-based data-driven ROM, into an error caused by the imperfect recollection of the parameter spaces Θ and Θ', an error caused by the imperfect projection given by the DOD or POD respectively and an error of imperfect attribution of latent dynamics on that projection space by a neural network. This seperation can be achieved in a similar manner for both types of algorithms.

Before we start with the formulation and proof of these decompositions, we want to predefine some elements of the corresponding theorems, to make them less cumbersome to read.

Firstly let in general n, N, N_A and N_h be the fixed natural number hierarchy from Assumption (5). Let the approximate POD-based parameter-to-solution map be defined as follows

$$\mathcal{G}_{\text{POD+DNN}} : \Theta \times \Theta' \times [0,T] \to \mathbb{R}^{N_h}; \quad (\mu,\nu,t) \mapsto \mathbb{A}_{\text{P}} \cdot \hat{q}_{\text{P}}(\mu,\nu,t),$$

where we let $\mathbb{A}_{\text{P}} \in \mathbb{R}^{N_h \times N}$ be a POD computed by Algorithm (1) for the dimension N with the mass matrix $\mathbb{G}$, and

$$\hat{q}_{\text{P}} := R_\rho(\Phi_N^{\text{P}}(\theta_{\text{PDNN}}))$$

be the realization of some neural network Φ_N^{P} with input dimension $p+q+1$ and output dimension N, and an already fixed weight $\theta_{\text{PDNN}} \in \Theta(\Phi_N^{\text{P}})$. And let similarily the approximate DOD-based parameter-to-solution map be defined as follows

$$\mathcal{G}_{\text{DOD+DNN}} : \Theta \times \Theta' \times [0,T] \to \mathbb{R}^{N_h}; \quad (\mu,\nu,t) \mapsto \mathbb{V}_{\mu,t} \cdot \hat{q}_{\text{D}}(\mu,\nu,t),$$

where we let

$$\mathbb{V}_{\mu,t} := \mathbb{A} \cdot \widetilde{\mathbb{V}}_{\mu,t}(\theta_{\text{DOD}}),$$

for an a priori fixed parameter $\theta_{\text{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ according to Definition (3.6) and the used POD matrix $\mathbb{A} \in \mathbb{R}^{N_h \times N_A}$, and

$$\hat{q}_{\text{D}} := R_\rho(\Phi_N(\theta_{\text{DDNN}}))$$

be the realization of some neural network Φ_N^{D} with input dimension $p+q+1$ and output dimension N, and an already fixed weight $\theta_{\text{DDNN}} \in \Theta(\Phi_N^{\text{D}})$.

Moreover we assume for this section to have after reordering and with $N_{\text{data}} = N_{s_1} N_{s_2} N_t$

$$\{(\mu_j, \nu_j, t_j), u_j\}_{j=1}^{N_{\text{data}}} := \left\{(\mu_i, \nu_j, k\Delta t), u_h^{\mu_i,\nu_j,k\Delta t}\right\}_{(i,j,k)=(1,1,1)}^{(N_{s_1}, N_{s_2}, N_t)} \subset (\mathcal{P}_1 \times \mathcal{P}_2 \times \mathcal{T}) \times \mathbb{R}^{N_h}$$

a set of training data conforming to the label and feature notation in Definition (2.32), where the labels have been sampled in accordance to Assumption (2).

For both algorithms, the relative error can be given as

$$\mathcal{E}_{\text{R}} := \left(\int_{\Theta \times \Theta' \times [0,T]} \frac{\|u_h^{\mu,\nu,t} - \hat{u}_h^{\mu,\nu,t}\|}{\|u_h^{\mu,\nu,t}\|} d(\mu, \nu, t) \right)^{1/2}, \tag{4.1}$$

where $\hat{u}_h^{\mu,\nu,t} \in \mathbb{R}^{N_h}$ can be given either with regard to the approximate parameter-to-solution map of the DOD+DNN or the POD+DNN.

Theorem 4.3 (POD+DNN Decomposition)**.** *Define the preliminaries such as in the introduction to this Section (4.1). Then under the Assumptions (2) and (3), we have for the relative error of the POD+DNN*

$$\mathcal{E}_R^{POD+DNN} \leq \mathcal{E}_S + \mathcal{E}_{POD} + \mathcal{E}_{NN},$$

where

1. *The sampling error*
$$\mathcal{E}_S = \mathcal{E}_S \left(\mathcal{G}, (\mu_i, \nu_i, t_i)_{i=1}^{N_{data}}, N \right)$$
satisfies $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$, with expectation
$$\mathbb{E}[\mathcal{E}_S] = \mathcal{O}(N_s^{-1/4} + N_t^{-1}).$$

2. *The POD projection error*
$$\mathcal{E}_{POD} = \mathcal{E}_{POD} \left(\mathcal{G}, (\mu_i, \nu_i, t_i)_{i=1}^{N_{data}}, N \right)$$
satisfies $\mathcal{E}_{POD} \to \mathcal{E}_{POD,\infty}$ almost surely as $N_s, N_t \to \infty$, where
$$\mathcal{E}_{POD,\infty} = \mathcal{E}_{POD,\infty}(\mathcal{G}, N)$$
is independent of the data snapshots and proportional to the linear KnW of the manifold $\mathcal{S} = \{u_h^{\mu,\nu,t} \mid (\mu, \nu, t) \in \Theta' \times [0, T]\}$.

3. *The neural network approximation error*
$$\mathcal{E}_{NN} = \mathcal{E}_{NN} \left(\mathcal{G}, N, \theta_{PDNN}, \Phi_N^P \right)$$
is arbitrarily small depending on the size and choice of weights of the neural network Φ_N^P.

Proof. Let

$$\|\cdot\|_{L^2_\omega} := \left(\int_{\Theta \times \Theta \times [0,T]} \frac{\|\cdot\|^2}{\|u_h^{\mu,\nu,t}\|^2} d(\mu, \nu, t) \right)^{1/2},$$

define a map $\|\cdot\|_{L^2_\omega} : L^2(\Theta \times \Theta' \times [0,T]; \mathbb{R}^{N_h}) \to \mathbb{R}^+$. Then this is a norm. We coarsely proof this, by remarking the inheritance of the triangle inequality from the $\|\cdot\|$ under the usage of the Minkowski inequality, i.e. for arbitrary $v, w \in L^2(\Theta \times \Theta' \times [0,T]; \mathbb{R}^{N_h})$

$$\begin{aligned}\|v+w\|^2_{L^2_\omega} &= \int_{\Theta\times\Theta'\times[0,T]} \frac{\|v(\mu,\nu,t)+w(\mu,\nu,t)\|^2}{\|u_h^{\mu,\nu,t}\|^2} d(\mu,\nu,t) \\ &\leq \int_{\Theta\times\Theta'\times[0,T]} \frac{\|v(\mu,\nu,t)\|^2+\|w(\mu,\nu,t)\|^2}{\|u_h^{\mu,\nu,t}\|^2} d(\mu,\nu,t) = \|v\|^2_{L^2_\omega} + \|w\|^2_{L^2_\omega},\end{aligned}$$

and the absolut homogeneity from simple qualities of the integral. Finally let $\|v\|_{L^2_\omega} = 0$ for some $v \in L^2(\Theta \times \Theta' \times [0,T])$, then assuming $v \neq 0$ yields with Assumption (3)

$$\|v\|^2_{L^2_\omega} \geq m^{-1} \int_{\Theta\times\Theta'\times[0,T]} \|v(\mu,\nu,t)\| d(\mu,\nu,t) > 0,$$

hence reaching a contradiction. With the knowledge of $\|\cdot\|_{L^2_\omega}$ being a norm, we know it must be subject to the triangle inequality, thus

$$\mathcal{E}_{\mathrm{R}} = \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\|u_h^{\mu,\nu,t} - \mathbb{A}\hat{q}_{\mathrm{P}}(\mu,\nu,t)\|}{\|u_h^{\mu,\nu,t}\|} d(\mu,\nu,t) \right)^{1/2} \tag{4.2}$$

$$\leq \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|_{L^2_\omega} + \|\mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t} - \mathbb{A}\hat{q}_{\mathrm{P}}(\mu,\nu,t)\|_{L^2_\omega}. \tag{4.3}$$

Let $q_A(\mu,\nu,t) := \mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}$ be a reduced solution according to the POD basis. Then the natural definition of the neural network approximation error is

$$\mathcal{E}_{\mathrm{NN}} = \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\|\mathbb{A}q_A(\mu,\nu,t) - \mathbb{A}\hat{q}(\mu,\nu,t)\|^2}{\|u_h^{\mu,\nu,t}\|^2} d(\mu,\nu,t) \right)^{1/2}. \tag{4.4}$$

This marks the second part in (4.3). This is only dependent on the ability of the neural network to identifying the best choice on a given linear sub-manifold of the solution manifold. The first part is concerned with the error generated from the best *possible* approximation of such a linear sub-manifold.

For this, we can bound the norm $\|\cdot\|_{L^2_\omega} \leq m^{-1}\|\cdot\|$, where $m > 0$ is the constant stemming from Assumption (3). Further we remind ourselves of the definition of the discrete correlation matrix $K = |\Theta \times \Theta \times \mathcal{T}| N_{data}^{-1} U\mathbb{G}U^T$ and its eigenvalues σ_k^2 in (2.10). Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$, we get

$$\begin{aligned}&\|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|_{L^2_\omega} \\ &\leq m^{-1} \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t) \right)^{1/2} \\ &\leq m^{-1} \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 + \sum_{k>N} \sigma_k^2 \right)^{1/2} \\ &\leq m^{-1} \left(\left| \int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 \right| + \sum_{k>N} \sigma_k^2 \right)^{1/2} \\ &\leq \underbrace{m^{-1} \left| \int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 \right|^{1/2}}_{=:\mathcal{E}_{\mathrm{S}}} + \underbrace{m^{-1} \sqrt{\sum_{k>N} \sigma_k^2}}_{=:\mathcal{E}_{\mathrm{POD}}}.\end{aligned}$$

This concludes the estimate up to the facts about the convergence of the errors. Recalling Proposition (2.12), we can rewrite the sampling error as follows

$$\mathcal{E}_{\mathrm{S}} = m^{-1}\left| \int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 \, d(\mu,\nu,t) - \frac{|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}} \sum_{j=1}^{N_{data}} \|u_j - \mathbb{A}\mathbb{A}^T\mathbb{G}u_j\|^2 \right|^{1/2}$$

Furthermore, we define, using the compactness postulated in Assumption (2) and the boundedness in Assumption (3), the L^2-error map via

$$f(\mu,\nu,t) := \|u_h^{\mu,\nu,t} - \mathbb{A}\mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 \leq M^2\|I_{N_h} - \mathbb{A}\mathbb{A}^T\mathbb{G}\|_{\mathrm{op}}^2 = M^2\|I_{N_A} - \mathbb{A}\|_{\mathrm{op}} < \infty.$$

Thus with $f \in L^2(\Theta\times\Theta'\times\mathcal{T})$, we know that by Proposition (4.2), $\mathbb{E}[\mathcal{E}_{\mathrm{S}}] \leq \mathcal{O}(N_s^{-1/4} + N_t^{-1/2})$.

By the strong law of large numbers, we can thus conclude, that $\mathcal{E}_{\mathrm{S}} \to 0$ almost surely, as $N_t, N_s \to \infty$, and further by Proposition (2.11), we have that $\mathcal{E}_{POD} \to \mathcal{E}_{POD,\infty}$ as $N_t, N_s \to \infty$, where

$$\mathcal{E}_{POD,\infty} = m^{-1}\sqrt{\sum_{k>N}\sigma_{k,\infty}^2}. \tag{4.5}$$

□

Even optimistically, the upper error bound decomposition of Theorem (4.3) suggest for an ever-increasing number of data points $N_s, N_t \to \infty$, and an ever-increasing complexity for the Deep Neural Network underlying the realization of $\hat{q}_{\mathrm{P}}$, an error of order of the Kolmogorov N-width of the solution manifold, since the DNN cannot overcome the unavoidable loss to even the perfect POD choice. Intuitively, the specific proportionality constant will be larger if the quotient between "small" regions of the solution space to the "large" regions of the solution space is big.

This observation can be reformulated into the following theorem.

Theorem 4.4. *Under the assumptions of Theorem (4.3), we have that*

$$\mathcal{E}_R^{POD+DNN} \geq \frac{m}{M}\mathcal{E}_{POD,\infty},$$

where $\mathcal{E}_{POD,\infty} := m^{-1}\sqrt{\sum_{k>N}\sigma_{k,\infty}^2}$.

Proof.

Proof this

□

We have now discussed some preliminary relative error results for the POD+DNN neural networks for given near-optimal neural network weights. As a next step, one can do the same for the DOD+DNN neural networks. First, we need an auxilary statement, that precidents the similarity of definition of the DOD projection error to the POD projection error, since the former depends on architecture.

The key idea here being, that one can employ Horniks Theorem to ensure existence of a neural network architecture and corresponding neural network of such architecture, such that the objective functional

$$(\mu,t,\mathbb{V}) \mapsto \int_{\Theta'} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|d(\nu),$$

is close to the "perfect one". Under triangle equality for the distance to the "perfect one", this can be seperated into the mistake, which is arbitrarily small, and the total size of the perfect objective functional, that is nothing but the Kolmogorov N-width under fixed (μ, t). And that KnW decays exponentially fast by Assumption (1), in contrast to the general one.

Remark 4.5. Generally we can *not* employ Yarotsky Theorem (2.30) here, since the optimal objective functional, as described above, can only be shown to be Lipschitz continous under our current assumptions, as given above, and Yarotsky does require C^2-smoothness. However, we will set up the proof for the following lemma in such a way, that Yarotsky can be directly interchanged, if a higher regularity for the optimal objective functional is imposed.

The source for the following lemma is generally (Franco et al., 2024).

Lemma 4.6 (Optimal Deep Orthogonal Decomposition)**.** For each geometric parameter $\mu \in \Theta$, let $\mathcal{S}_{\mu,t} := \{u_h^{\mu,\nu,t}\}_{\nu\in\Theta'} \subset \mathcal{G}(\Theta \times \Theta' \times [0, T])$ be the (μ, t)-invariant solution submanifold. Then, under the Assumptions (2) and (3), for every $\varepsilon > 0$ there are weights $\theta^*_{\text{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ for a DOD of output dimension $N_h \times N$ accoring to Definition (3.6) such that

$$\int_{\Theta\times\Theta'\times[0,T]} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}(\theta^*_{\text{DOD}})\mathbb{V}_{\mu,t}(\theta^*_{\text{DOD}})\mathbb{G}u_h^{\mu,\nu,t}\| d(\mu,\nu,t) < \varepsilon + \int_{\Theta\times[0,T]} d_N(\mathcal{S}_{\mu,t}) d(\mu, t).$$

Proof. Let $N, N_A, N_h \in \mathbb{N}$ be fixed by Assumption (5). Assuming we have $\mu \in \Theta, t \in [0, T]$ and $\mathbb{V} = \mathbb{A}\widetilde{\mathbb{V}} \in [0, 1]^{N_h\times N}$, with $\mathbb{A} \in [0, 1]^{N_h\times N_A}$ being $\mathbb{G}$ orthonormal, and $\widetilde{\mathbb{V}} \in [0, 1]^{N_A\times N}$, then we define $J : \Theta \times [0, T] \times [0, 1]^{N_h\times N} \to \mathbb{R}$ via

$$J(\mu, t, \mathbb{V}) := \int_{\Theta'} \|u_h^{\mu,\nu,t} - \mathbb{V}\mathbb{V}^T\mathbb{G}u_h^{\mu,\nu,t}\| d(\nu) = \int_{\Theta'} |u_{\text{pre-red}}^{\mu,\nu,t} - \widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T u_{\text{pre-red}}^{\mu,\nu,t}| d(\nu),$$

for $u_{\text{pre-red}}^{\mu,\nu,t} := \mathbb{A}^T\mathbb{G}u_h^{\mu,\nu,t} \in \mathbb{R}^{N_A}$ for each $\mu \in \Theta, \nu \in \Theta'$ and $t \in [0, T]$, and let some $\varepsilon > 0$ be fixed. Under Assumption (3), one can infer the Lipschitz continuity of J by the Lipschitz continuity of $\mathcal{G}$, i.e. let $\mu, \mu' \in \Theta$, $t, t' \in [0, T]$ and $\mathbb{V} = \mathbb{A}\widetilde{\mathbb{V}}, \mathbb{V}' = \mathbb{A}\widetilde{\mathbb{V}}' \in \mathbb{R}^{N_h\times N}$ such that for some $\delta > 0$

$$|\mu - \mu'| + |t - t'| + \|\mathbb{V} - \mathbb{V}'\|_{\text{op}} < \delta.$$

For the following calculation, we compose the preliminary identity for $u := u_{\text{pre-red}}^{\mu,\nu,t}$ and $u' := u_{\text{pre-red}}^{\mu',\nu,t'}$, and the projectors $P := \widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T$, P' analogously:

$$\begin{aligned}
(\mathbb{I}_{N_A} - P)u - (\mathbb{I}_{N_A} - P')u' &= u - Pu - u' + P'u' \\
&= u - Pu - u' + P'u' + Pu' - Pu' \\
&= u - u' - Pu + Pu' + P'u' - Pu' \\
&= (\mathbb{I}_{N_A} - P)(u - u') + (P' - P)u'.
\end{aligned} \tag{4.6}$$

Then we can calculate with, the reverse triangle inequality, and our auxiliary Equation (4.6),

$$\begin{aligned}
|J(\mu, t, \mathbb{V}) - J(\mu', t', \mathbb{V}')| &= \int_{\Theta'} |\|u_h^{\mu,\nu,t} - \mathbb{V}\mathbb{V}^T\mathbb{G}u_h^{\mu,\nu,t}\| - \|u_h^{\mu',\nu,t} - \mathbb{V}'(\mathbb{V}')^T\mathbb{G}u_h^{\mu',\nu,t}\|| d(\nu) \\
&= \int_{\Theta'} \left| |u_{\text{pre-red}}^{\mu,\nu,t} - \widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T u_{\text{pre-red}}^{\mu,\nu,t}| - |u_{\text{pre-red}}^{\mu',\nu,t'} - \widetilde{\mathbb{V}}'(\widetilde{\mathbb{V}}')^T u_{\text{pre-red}}^{\mu',\nu,t'}| \right| d(\nu) \\
&= \int_{\Theta'} \left| \left(u_{\text{pre-red}}^{\mu,\nu,t} - \widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T u_{\text{pre-red}}^{\mu,\nu,t}\right) - \left(u_{\text{pre-red}}^{\mu',\nu,t'} - \widetilde{\mathbb{V}}'(\widetilde{\mathbb{V}}')^T u_{\text{pre-red}}^{\mu',\nu,t'}\right) \right| d(\nu) \\
&\leq \int_{\Theta'} \left|(\mathbb{I}_{N_A} - \widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T)(u_{\text{pre-red}}^{\mu,\nu,t} - u_{\text{pre-red}}^{\mu',\nu,t'})\right| d(\nu) + M|\Theta'|\|\widetilde{\mathbb{V}}\widetilde{\mathbb{V}}^T - \widetilde{\mathbb{V}}'(\widetilde{\mathbb{V}}')^T\|_{\text{op}} \\
&< \delta|\Theta'|(L + 2M),
\end{aligned}$$

and is thus Lipschitz continous aswell.

big step here

Hence there is a Borel measurable map $s : \Theta \times [0,T] \to [0,1]^{N_A \times N}$ producing w.l.o.g. orthonormal matrices given by

$$s : (\mu, t) \mapsto \underset{\widetilde{\mathbb{V}} \in [0,1]^{N_A \times N}}{\operatorname{argmin}} \ J(\mu, t, \mathbb{A}\widetilde{\mathbb{V}}).$$

This map is bounded (given by continuity of J and compactness of its image) and therefore $s \in L^1(\Theta \times [0,T]; \mathbb{R}^{N_A \times N})$, i.e.

$$\|s\|_{L^1(\Theta \times [0,T]; \mathbb{R}^{N_A \times N})} := \int_{\Theta \times [0,T]} \|s(\mu, t)\|_{\text{op}} \, d(\mu, t) < \infty.$$

Define $[s]_{ij}$ as the canoncially lifted entry in position $(i,j) \in \{1, \ldots, N_A\} \times \{1, \ldots, N\}$.

This means that by Hornik Theorem (2.20), there is a neural network architecture $\mathcal{A}^{[\widetilde{\mathbb{V}}_0]_{ij}}$ with $(p+1)$-dimensional input and one-dimensional ouput, and a neural network of that architecture $[\widetilde{\mathbb{V}}_0]_{ij}$, satisfying

$$\|[s]_{ij} - R_\rho([\widetilde{\mathbb{V}}_0]_{ij})\|_{L^1(\Theta \times [0,T]; \mathbb{R})} \leq \varepsilon,$$

for each $(i,j) \in \{1, \ldots, N_A\} \times \{1, \ldots, N\}$, and then using network parallelization as in Definition (2.24), we can set the optimal DOD neural network as

$$\widetilde{\mathbb{V}}_0 := P\left(P([\widetilde{\mathbb{V}}_0]_{11}, \ldots, [\widetilde{\mathbb{V}}_0]_{1N}), \ldots, P([\widetilde{\mathbb{V}}_0]_{N_A 1}, \ldots, [\widetilde{\mathbb{V}}_0]_{N_A N}) \right),$$

since it attains the wanted accuracy

$$\|s - R_\rho(\widetilde{\mathbb{V}}_0)\|_{L^1(\Theta \times [0,T]; \mathbb{R}^{N_A \times N})} \leq \varepsilon.$$

To now ensure, that the provided realization of the DOD neural network are indeed in the pre-image of J, we need to ensure, that there is a network, which indeed attains the accuracy, but is also entrywise $[0,1]$. For this, let $L = ((A_1, b_1), (A_2, b_2))$ be given via

$$A_1 := \begin{bmatrix} \mathbb{I}_{\mathbb{R}^{N_A \times N}} \\ \mathbb{I}_{\mathbb{R}^{N_A \times N}} \end{bmatrix}, \quad b_1 := \begin{pmatrix} 0_{\mathbb{R}^{N_A \times N}} \\ -\mathbb{I}_{\mathbb{R}^{N_A \times N}} \end{pmatrix}, \quad A_2 := \left[\mathbb{I}_{\mathbb{R}^{N_A \times N}} \;\middle|\; -\mathbb{I}_{\mathbb{R}^{N_A \times N}} \right], \quad b_2 := 0.$$

The realization of this network is equivalent to the entry-wise operator of $x \mapsto \rho(x) - \rho(x-1) \in [0,1]$ for $x \in \mathbb{R}$. Since this realization as a map is bounded by 1, we can infer with the definition $\mathbb{V}_{\mu,t} := R_\rho(L(\theta^*_{\text{DOD}})) \circ (\mathbb{A} \cdot R_\rho(\widetilde{\mathbb{V}}_0(\theta^*_{\text{DOD}})))(\mu, t)$ a new network with a final architecture $\Phi_{\widetilde{\mathbb{V}}}$ and complexity only differing by constants, and a choice of neural network weights $\theta^*_{\text{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$, such that

$$\int_{\Theta \times [0,T]} \|\mathbb{V}_{\mu,t} - s(\mu,t)\|_{\text{op}} d(\mu,t) = \int_{\Theta \times [0,T]} \underbrace{\|L\|_{\text{op}}}_{\leq 1} \|R_\rho(\widetilde{\mathbb{V}}_0)(\mu,t) - s(\mu,t)\|_{\text{op}} d(\mu,t) < \varepsilon.$$

Finally, with the knowledge, that $\mathbb{V}_{\mu,t}$ is ε-close to $s(\mu,t)$ in average, and the fact that this, as a matrix, is indeed in the pre-image of J, we can write using the Lipschitz continuity of J

$$\begin{aligned}\int_{\Theta\times\Theta'\times[0,T]} &\|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\| d(\mu,\nu,t) \\ &\leq \int_{\Theta\times[0,T]} |J(\mu,t,\mathbb{V}_{\mu,t}) - J(\mu,t,s(\mu,t))|\, d(\mu,t) + \int_{\Theta\times[0,T]} J(\mu,t,s(\mu,t)) d(\mu,t) \\ &< \varepsilon + \int_{\Theta\times[0,T]} \left(\min_{\mathbb{V}\in\mathbb{R}^{N_h\times N}} J(\mu,t,\mathbb{V})\right) d(\mu,t) \\ &< \varepsilon + \int_{\Theta\times[0,T]} \left(\min_{\mathbb{V}\in\mathbb{R}^{N_h\times N}} \sup_{\nu\in\Theta'} \|u_h^{\mu,\nu,t} - \mathbb{V}\mathbb{V}^T\mathbb{G}u_h^{\mu,\nu,t}\|\right) d(\mu,t).\end{aligned}$$

□

Theorem 4.7. *Define the preliminaries such as in the introduction to this Section (4.1). Further set $\mathbb{V}_{\mu,t} := \mathbb{V}_{\mu,t}(\theta^*_{DOD})$ for the choice of weights $\theta^*_{DOD} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ as in Lemma (4.6). Then, under the Assumptions (2) and (3), we have for the relative error of the DOD+DNN*

$$\mathcal{E}_R^{DOD+DNN} \leq \mathcal{E}_S + \mathcal{E}_{DOD} + \mathcal{E}_{NN}, \tag{4.7}$$

where

1. *The sampling error*

$$\mathcal{E}_S = \mathcal{E}_S\left(\mathcal{G}, (\mu_i,\nu_i,\tau_i)_{i=1}^{N_{data}}, N\right)$$

satisfies $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$, with expectation

$$\mathbb{E}[\mathcal{E}_S] = \mathcal{O}(N_s^{-1/4} + N_t^{-1}).$$

2. *The DOD projection error*

$$\mathcal{E}_{DOD} = \mathcal{E}_{DOD}\left(\mathcal{G}, (\mu_i,\nu_i,\tau_i)_{i=1}^{N_{data}}, N\right)$$

satisfies $\mathcal{E}_{DOD} \to \mathcal{E}_{DOD,\infty}$ almost surely as $N_s, N_t \to \infty$, where

$$\mathcal{E}_{DOD,\infty} = \mathcal{E}_{DOD,\infty}(\mathcal{G}, N)$$

is independent of the data snapshots and arbitrarily close, with regards to the size of the DOD neural network, to the order of the averaged linear KnW of the solution submanifold $\mathcal{S}_{\mu,t} = \{u_h^{\mu,\nu,t} \,|\, \nu \in \Theta'\}$.

3. *The neural network approximation error*

$$\mathcal{E}_{NN} = \mathcal{E}_{NN}(\mathcal{G}, N, \hat{q})$$

is arbitrarily small depending on the size and choice of weights of the neural network Φ_A.

Proof. In its core, this proof inherits its structure and arguments from the one of Theorem (4.3), we will therefore only shorten the parts in which they are interchangeable. Let $\|\cdot\|_{L^2_\omega} : L^2(\Theta\times\Theta'\times[0,T];\mathbb{R}^{N_h}) \to \mathbb{R}^+$ be the norm as given before, and define

$$q_N(\mu,\nu,t) := \mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t} \in \mathbb{R}^N,$$

as the reduced solution regarding the DOD projection. Then we can apply the triangle inequality like before, to achieve

$$\mathcal{E}_{\mathrm{R}}^{\mathrm{DOD+DNN}} \leq \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|_{L^2_\omega} + \|\mathbb{V}_{\mu,t}q_N(\mu,\nu,t) - \mathbb{V}_{\mu,t}\hat{q}_{\mathrm{D}}(\mu,\nu,t)\|_{L^2_\omega}.$$

In turn, from here we can already infer the error stemming from the latent dynamics identification under the neural network realization $\hat{q}_{\mathrm{D}}$, i.e.

$$\mathcal{E}_{NN} = \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\|\mathbb{V}_{\mu,t}q_N(\mu,\nu,t) - \mathbb{V}_{\mu,t}\hat{q}_{\mathrm{D}}(\mu,\nu,t)\|^2}{\|u_h^{\mu,\nu,t}\|^2} d(\mu,\nu,t)\right)^{1/2}. \tag{4.8}$$

Moreover, we can extrapolate the same procedure of splitting the remaining term into the respective errors, where we already assume for the fixed $N \in \mathbb{N}$, the "optimal" choice of weights $\theta^*_{\mathrm{DOD}} \in \Theta(\Phi_{\widetilde{\mathbb{V}}})$ for the underlying DOD neural network in accordance to Lemma (4.6)

$$\|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|_{L^2_\omega} \leq \mathcal{E}_{\mathrm{S}} + \mathcal{E}_{\mathrm{DOD}}, \tag{4.9}$$

where

$$\begin{aligned}\mathcal{E}_S = m^{-1}\Bigg| &\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t)\\ &- \frac{|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}} \sum_{i=1}^{N_{s_1}}\sum_{j=1}^{N_{s_2}}\sum_{k=1}^{N_t} \|u_h^{\mu_i,\nu_j,k\Delta t} - \mathbb{V}_{\mu_i,k\Delta t}\mathbb{V}_{\mu_i,k\Delta t}^T\mathbb{G}u_h^{\mu_i,\nu_j,k\Delta t}\|^2\Bigg|^{1/2}\end{aligned}$$

and

$$\mathcal{E}_{\mathrm{DOD}} = \frac{m^{-1}|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}}\Bigg| \sum_{i=1}^{N_{s_1}}\sum_{j=1}^{N_{s_2}}\sum_{k=1}^{N_t} \|u_h^{\mu_i,\nu_j,k\Delta t} - \mathbb{V}_{\mu_i,k\Delta t}\mathbb{V}_{\mu_i,k\Delta t}^T\mathbb{G}u_h^{\mu_i,\nu_j,k\Delta t}\|^2\Bigg|^{1/2}.$$

This leaves us with the proof for the convergence statements. While $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$ can be shown in the same manner as it has been in the aforementioned proof above, the DOD error is more tricky.

Using the triangle inequality again and combining it with the findings of Lemma (4.6) for the lemma-specific $\varepsilon > 0$

$$\begin{aligned}\mathcal{E}_{\mathrm{DOD}} &= m^{-1}\Bigg|\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t)\Bigg|^{1/2}\\ &\quad + \mathcal{E}_{\mathrm{DOD}} - m^{-1}\Bigg|\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t)\Bigg|^{1/2}\\ &\xrightarrow[N_s,N_t\to\infty]{} \underbrace{m^{-1}\Bigg|\int_{\Theta\times\Theta'\times\mathcal{T}} \|u_h^{\mu,\nu,t} - \mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T\mathbb{G}u_h^{\mu,\nu,t}\|^2 d(\mu,\nu,t)\Bigg|^{1/2}}_{=:\mathcal{E}_{\mathrm{DOD},\infty}}\\ &\quad < m^{-1}\varepsilon + m^{-1}\int_{\Theta\times\mathcal{T}} d_N(\mathcal{S}_{\mu,t})d(\mu,t),\end{aligned}$$

which can be arbitrarily improved for a more complex choice of the inner DOD module architecture. This dependency on the complexity is suggested by Theorem (2.30), since the L^∞ error correlates with the $\mathcal{H}_h$ representation error by the compactness of the time-parameter space, that is given by Assumption (2). □

4.2 Upper Complexity Bounds

Theorem 4.8. *Let $\mathcal{G} : (\mu, \nu, t) \mapsto u_h^{\mu,\nu,t}$ for any $(\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}$ be the parameter-to-solution map and let Assumption (2) and (3) be true. Let $1 > \delta > 0$ and $\varepsilon > 0$. Now assume we can collect freely a sufficient amount of training data $N_{data} = N_{data}(\delta, \varepsilon)$ and put it into the correlation matrix K as given in Definition (2.10) with σ_k^2 being its sorted eigenvalues. Now choose*

$$N = \operatorname{argmin} \left\{ j \in \mathbb{N} \,\middle|\, \sum_{k>j} \sigma_k^2 \leq \frac{m^2}{9} \varepsilon^2 \right\}.$$

Given the POD matrix $V \in \mathbb{R}^{N_h \times N}$ we now define the parameter-to-POD-coefficient map $\mathcal{Q} : (\mu, \nu, t) \mapsto q(\mu, \nu, t) := V^T u_h^{\mu,\nu,t}$ for any $(\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}$. We assume that there exists $n > 0, \psi_* : \mathbb{R}^n \to \mathbb{R}^N, \psi'_* : \mathbb{R}^N \to \mathbb{R}^n$ that are respectively s-times and s'-times differentiable (with $s >> s' \geq 2$), such that they enjoy the perfect embedding Assumption (4), i.e.

$$\psi_*(\psi'_*(q(\mu, \nu, t))) = q(\mu, \nu, t), \quad \forall (\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}.$$

We further let

$$C_1 := \sup_{|\alpha| \leq s'} \sup_{v \in \mathbb{R}^N} |\, D^\alpha \psi'_*(v) \,|, \quad C_2 := \sup_{|\alpha| \leq s} \sup_{w \in \mathbb{R}^n} |\, D^\alpha \psi'_*(w) \,|.$$

Then there is a constant $c = c(\Theta, \Theta', \mathcal{T}, L, C_1, C_2, p, q, n, s, s')$ and a POD-DL-ROM architecture $V\hat{q} = V\psi \circ \phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^{N_h}$ composed of a decoder $\psi : \mathbb{R}^n \to \mathbb{R}^N$ having at most:

- $L_{n \to N} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{n \to N} = cN\varepsilon^{-n/(s-1)} \log(\varepsilon^{-1})$ *active weights,*

and a reduced feed forward neural network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$ having at most:

- $L_{(p+q+1) \to n} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{(p+q+1) \to n} = cn\varepsilon^{-(p+q+1)} \log(\varepsilon^{-1})$ *active weights,*

such that $\mathbb{P}[\mathcal{E}_R < \varepsilon] > 1 - \delta$.

Proof. We can directly bound $\mathcal{E}_S = \mathcal{E}_S(N_{s_1}, N_{s_2}, N_t)$ independently of N, under the Assumption (2) by the Weak Law of Large Numbers, i.e. for all $1 > \delta > 0$ and $\varepsilon > 0$ there are N_{s_1}, N_{s_2}, N_t, such that

$$\mathbb{P}[\mathcal{E}_S((N_{s_1}, N_{s_2}, N_t)) < \varepsilon/3] > 1 - \delta. \tag{4.1}$$

This result provides the "probability" part of the PAC (probably almost correct) bound, which we want to show. Bounding $\mathcal{E}_{\text{NN}}$ and $\mathcal{E}_{\text{POD}}$ by $\varepsilon/3$ respectively will then provide the wanted result. For this, by choosing N as in the assumption of the statement, we simply rewrite

$$\mathcal{E}_{\text{POD}} := m^{-1} \sqrt{\sum_{k>N} \sigma_k^2} \leq \frac{\varepsilon}{3}. \tag{4.2}$$

To now bound the neural network approximation error, we will need to simply rewrite the error in terms of only the norm of the network itself;

$$
\begin{aligned}
\mathcal{E}_{\mathrm{NN}} &= \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\|Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2}{\|u_h^{\mu,\nu,t}\|^2} \, d(\mu,\nu,t) \right)^{1/2} \\
&\leq m^{-1} \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \|Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2 \, d(\mu,\nu,t) \right)^{1/2} \\
&\leq m^{-1} \left(|\Theta\times\Theta'\times\mathcal{T}| \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2 \right)^{1/2} \\
&\leq m^{-1} \left(|\Theta\times\Theta'\times\mathcal{T}| \, \|V\|_\infty^2 \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|q(\mu,\nu,t) - \hat{q}(\mu,\nu,t)\|^2 \right)^{1/2} \\
&= m^{-1} \, |\Theta\times\Theta'\times\mathcal{T}|^{1/2} \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|q(\mu,\nu,t) - \hat{q}(\mu,\nu,t)\|.
\end{aligned}
\tag{4.3}
$$

whereas we split the latter error into both parts, since $\hat{q} = \psi\circ\phi$ per definition. This means employing two types of error estimations in the matter of finding the respective optimal maps for the feed forward estimation of the latent dynamics via $\phi_* : \mathbb{R}^{(p+q+1)} \to \mathbb{R}^n$ and the decoder lift via $\psi_* : \mathbb{R}^n \to \mathbb{R}^N$.

- Let $\mathcal{S}_N := \{q = \mathcal{Q}(\mu,\nu,t) \,|\, (\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}\}$, then the image of the optimal embedding $\mathcal{V}_n = \psi_*'(\mathcal{S}_N)$ is such that $\mathrm{diam}(\mathcal{V}_N) \leq LC_1\mathrm{diam}(\Theta\times\Theta'\times\mathcal{T})$, given the Assumption (3) by continuity.

  Make this rigorous

  Thus, by Theorem Gühring, there is a ReLU Neural Network $\psi : \mathbb{R}^n \to \mathbb{R}^N$, such that

$$
\sup_{v\in\mathcal{V}_N} \|\psi(v) - \psi_*(v)\| < \frac{m}{6} |\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.4}
$$

$$
\operatorname*{ess\,sup}_{v,v'\in\mathcal{V}_N} \frac{|\,(\psi-\psi_*)(v) - (\psi-\psi_*)(v')\,|}{|\,v - v'\,|} < \frac{m}{6} |\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.5}
$$

  with $L_{n\to N} = c\log(\varepsilon^{-1})$ layers and $\omega_{n\to N} = cN\varepsilon^{-n/(s-1)}\log(\varepsilon^{-1})$ active weights.

  Make this rigorous

- Now let $\phi_*(\mu,\nu,t) = \psi_*'(q(\mu,\nu,t))$ for $(\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}$, which is Lipschitz continuous, with constant LC_1, and thus, by

  Make this rigorous

  Theorem Yarotski, there exists a ReLU Neural Network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$, such that

$$
\sup_{(\mu,\nu,t)\in\Theta\times\Theta\times\mathcal{T}} \|\phi(\mu,\nu,t) - \phi_*(\mu,\nu,t)\| < \frac{m}{6C_3} |\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.6}
$$

  with $L_{(p+q+1)\to n} = c\log(\varepsilon^{-1})$ layers and $\omega_{(p+q+1)\to n} = cn\varepsilon^{-(p+q+1)}\log(\varepsilon^{-1})$ active weights.

Starting from the last inequality in (4.3) we can perform the split in a clean fashion using the triangle inequality and reminding ourselves of the Assumption (4), which guarantees $q = \psi_*(\psi_*')$, and thus gives with both neural network estimates in Equations (4.4) and (4.6)

$$
\begin{aligned}
&\sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \|q(\mu,\nu,t)-\hat{q}(\mu,\nu,t)\| \\
&\leq \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \left(\|\psi_*(\psi_*'(\mu,\nu,t))-\psi(\phi_*(\mu,\nu,t))\| + \|\psi(\phi_*(\mu,\nu,t))-\psi(\phi(\mu,\nu,t))\| \right) \\
&\leq \sup_{v\in\mathcal{V}_N} \|\psi_*(v)-\psi(v)\| + C_3 \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \left(\|\psi(\phi_*(\mu,\nu,t))-\psi(\phi(\mu,\nu,t))\| \right) \\
&< \frac{m}{6}|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon + C_3\frac{m}{6C_3}|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon = \frac{\varepsilon}{3}\,m\,|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2},
\end{aligned}
$$

which cancels with $m^{-1}\,|\Theta\times\Theta'\times\mathcal{T}|^{1/2}$ to get

$$
\mathcal{E}_{\mathrm{NN}} < \frac{\varepsilon}{3}. \tag{4.7}
$$

Now putting together all three estimates (4.1), (4.2) and (4.7) with the reasoning of Theorem (4.3) finally gives

$$
\mathcal{E}_R \leq \mathcal{E}_S + \mathcal{E}_{\mathrm{POD}} + \mathcal{E}_{\mathrm{NN}} < \frac{\varepsilon}{3}+\frac{\varepsilon}{3}+\frac{\varepsilon}{3} = \varepsilon,
$$

with higher probability than $1-\delta$. □

Lemma 4.9 (Projection Error)**.** Let N_{s_2} be given under Assumption (2), then for all $\varepsilon > 0$ there is a ReLU neural network $V:\mathbb{R}^{p+1}\to\mathbb{R}^{N_h\times N}$, such that for $u_j := u(\mu,\nu_j,t)$ with given $\mu\in\Theta$ and $t\in\mathcal{T}$

$$
\sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \frac{1}{N_{s_2}}\sum_{j=1}^{N_{s_2}}\|u_j-\mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T u_j\|^2 - \min_{W\in\mathbb{R}^{N_h\times N}} \frac{1}{N_{s_2}}\sum_{j=1}^{N_{s_2}}\|u_j-WW^T u_j\|^2 \right|^{1/2} < \varepsilon.
$$

Proof. The existence and identity of V^* s.t.

$$
V^* = \operatorname*{argmin}_{W\in\mathbb{R}^{N_h\times N}} \sum_{j=1}^{N_{s_2}}\|u_j - V^*(V^*)^T u_j\|^2
$$

is given via the statement in Proposition (2.12) with the POD basis. Hence defining a norm via

$$
\|(u_j)_{j=1}^{N_{s_2}}\|_{train} := \left| \sum_{j=1}^{N_{s_2}}\|u_j\|^2 \right|^{1/2}
$$

gives us the following inequality firstly using the monotonicity of the square root and then using the triangle inequalities multiple times

$$
\begin{aligned}
&N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \sum_{j=1}^{N_{s_2}}\|u_j-\mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T u_j\|^2 - \sum_{j=1}^{N_{s_2}}\|u_j-V^*(V^*)^T u_j\|^2 \right|^{1/2} \\
&\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \sum_{j=1}^{N_{s_2}}\|\mathbb{V}_{\mu,t}\mathbb{V}_{\mu,t}^T u_j - V^*(V^*)^T u_j\|^2 \right|^{1/2},
\end{aligned}
$$

then define $q_j := (V^*)^T u_j$ and $\hat{q}_j := \mathbb{V}_{\mu,t}^T u_j$ and their training vectors accordingly, i.e. $q := (q_j)_{j=1}^{N_{s_2}}, \hat{q} := (\hat{q}_j)_{j=1}^{N_{s_2}}$

$$
\begin{aligned}
&\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|\mathbb{V}_{\mu,t}\hat{q} - \mathbb{V}_{\mu,t}q\|_{train} + \|\mathbb{V}_{\mu,t}q - V^*q\|_{train} \\
&\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|\mathbb{V}_{\mu,t}\|_\infty \|\hat{q} - q\|_{train} + \|\mathbb{V}_{\mu,t} - V^*\|_\infty \|q\|_{train} \\
&= N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|\mathbb{V}_{\mu,t}\|_\infty \|\mathbb{V}_{\mu,t}^T u - (V^*)^T u\|_{train} + \|\mathbb{V}_{\mu,t} - V^*\|_\infty \|(V^*)^T u\|_{train} \\
&\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|\mathbb{V}_{\mu,t}\|_\infty \|\mathbb{V}_{\mu,t} - V^*\|_\infty \|u\|_{train} + \|\mathbb{V}_{\mu,t} - V^*\|_\infty \|V^*\|_\infty \|u\|_{train} \\
&\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|u\|_{train} \left(\|\mathbb{V}_{\mu,t} - V^*\|_\infty \left(\|\mathbb{V}_{\mu,t}\|_\infty + 1\right)\right) \\
&\leq 2M \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \|\mathbb{V}_{\mu,t} - V^*\|_\infty,
\end{aligned}
$$

where we used $\|u\|_{train} \leq |\sum_{j=1}^{N_{s_2}} M^2|^{1/2} = N_{s_2}^{1/2} M$ and additionally $\|\mathbb{V}_{\mu,t}\|_\infty = \|V^*\|_\infty = 1$ by orthonormality in the last arguments.

Do the Yarotski argument with the appropriate number of layers and weights.

□

Extrapolate to DOD-DL-ROM

Theorem 4.10. *Let $\mathcal{G} : (\mu, \nu, t) \mapsto u_h^{\mu,\nu,t}$ for any $(\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}$ be the parameter-to-solution map and let Assumption (2) and (3) be true. Let $1 > \delta > 0$ and $\varepsilon > 0$. Now assume we can collect freely a sufficient amount of training data $\Theta_{data}, \Theta'_{data}$ and $\mathcal{T}_{data}$ of size $N_{data} = N_{data}(\delta, \varepsilon)$ and put all samples of $u(\mu, \nu_j, t)_{j=1}^{N_{s_2}}$ into a correlation matrix $K_{\mu,t}$ for each sample $(\mu, t) \in \Theta_{data} \times \mathcal{T}_{data}$ as given in Definition (2.10) with $\sigma_k(\mu, t)_k^2$ being its sorted eigenvalues. Now choose*

$$
N_{DOD} = \operatorname{argmin} \left\{ j \in \mathbb{N} \,\middle|\, \sup_{(\mu,t)\in\Theta_{data}\times\mathcal{T}_{data}} \sum_{k>j} \sigma(\mu, t)_k^2 \leq \frac{m^2}{16}\varepsilon^2 \right\}.
$$

For each pair $(\mu, t) \in \Theta_{data} \times \mathcal{T}_{data}$ given the respective POD matrix, so the best possible projection given the data set (refer to (2.12)), $V^*(\mu, t) \in \mathbb{R}^{N_h \times N_{DOD}}$ we now define the parameter-to-optimal-coefficient map $\mathcal{Q} : (\mu, \nu, t) \mapsto q(\mu, \nu, t) := V^*(\mu, t)^T u_h^{\mu,\nu,t}$ for any $(\mu, \nu, t) \in \Theta_{data} \times \Theta' \times \mathcal{T}_{data}$. We assume that there exists $n > 0, \Psi_* : \mathbb{R}^n \to \mathbb{R}^{N_{DOD}}, \Psi'_* : \mathbb{R}^{N_{DOD}} \to \mathbb{R}^n$ that are respectively s-times and s'-times differentiable (with $s >> s' \geq 2$), such that they enjoy the perfect embedding Assumption (4), i.e.

$$
\Psi_*(\Psi'_*(q(\mu, \nu, t))) = q(\mu, \nu, t), \quad \forall (\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}.
$$

We further let

$$
C'_1 := \sup_{|\alpha|\leq s'} \sup_{v\in\mathbb{R}^{N_{DOD}}} |\, D^\alpha \Psi'_*(v) \,|, \quad C'_2 := \sup_{|\alpha|\leq s} \sup_{w\in\mathbb{R}^n} |\, D^\alpha \Psi'_*(w) \,|.
$$

Then there is a constant $c = c(\Theta, \Theta', \mathcal{T}, L, C_1, C_2, p, q, n, s, s')$ *and a DOD-DL-ROM architecture* $V\hat{q} = V_{DOD}\Psi \circ \phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^{N_h}$ *composed of a decoder* $\Psi : \mathbb{R}^n \to \mathbb{R}^{N_{DOD}}$ *having at most:*

- $L_{n \to N_{DOD}} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{n \to N_{DOD}} = c\, N_{DOD}\, \varepsilon^{-n/(s-1)} \log(\varepsilon^{-1})$ *active weights,*

a reduced feed forward neural network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$ *having at most:*

- $L_{(p+q+1) \to n} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{(p+q+1) \to n} = cn\varepsilon^{-(p+q+1)} \log(\varepsilon^{-1})$ *active weights,*

and a DOD Module $V_{DOD} : \mathbb{R}^{p+1} \to \mathbb{R}^{N_h \times N_{DOD}}$ *having at most:*

- $L_{(p+1) \to N_h \times N_{DOD}} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{(p+1) \to N_h \times N_{DOD}} = c\, N_h\, N_{DOD}\, \varepsilon^{-(p+1)} \log(\varepsilon^{-1})$ *active weights,*

such that $\mathbb{P}[\mathcal{E}_R < \varepsilon] > 1 - \delta$*.*

Proof. Similar as to the proof of Theorem (4.8), we will try to bound each quantity of the statement in Theorem (4.7), but with the catch, that $\mathcal{E}_{\text{DOD}}$ is stochastic in nature, and thus we will employ Lemma (4.9) to find an auxilary term, which is independent of the data, to which the approximation of the neural net of the DOD is sufficiently "close".

The sampling error can be bounded in the same way as before, i.e. we can directly bound $\mathcal{E}_S = \mathcal{E}_S(N_{s_1}, N_{s_2}, N_t)$ independently of N_{DOD}, under the Assumption (2) by the Weak Law of Large Numbers, i.e. for all $1 > \delta > 0$ and $\varepsilon > 0$ there are N_{s_1}, N_{s_2}, N_t, such that

$$\mathbb{P}[\mathcal{E}_S((N_{s_1}, N_{s_2}, N_t)) < \varepsilon/4] > 1 - \delta. \tag{4.8}$$

Now recall the definition of $\mathcal{E}_{\text{DOD}}$ in the proof of Theorem (4.7) and further establish

$$\begin{aligned}
\mathcal{E}_{\text{DOD}} &= \frac{m^{-1}|\Theta \times \Theta' \times \mathcal{T}|}{N_{data}^{1/2}} \left| \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} \| u_h^{\mu_i,\nu_j,k\Delta t} - \mathbb{V}_{\mu_i,k\Delta t} \mathbb{V}_{\mu_i,k\Delta t}^T u_h^{\mu_i,\nu_j,k\Delta t} \|^2 \right|^{1/2} \\
&\le m^{-1} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \frac{|\Theta \times \Theta' \times \mathcal{T}|}{N_{s_2}} \sum_{j=1}^{N_{s_2}} \| u(\mu,\nu_j,t) - \mathbb{V}_{\mu,t} \mathbb{V}_{\mu,t}^T u(\mu,\nu_j,t) \|^2 \right|^{1/2},
\end{aligned}$$

enabling us to use Proposition (2.12) in order to switch to the eigenvalues $\sigma(\mu,t)_k$ for $(\mu,t) \in \Theta_{\text{data}} \times \mathcal{T}_{\text{data}}$ of the theoretical POD matrix $V^*(\mu,t)$ by employing the Lemma (4.9) with the bound

$$\frac{\varepsilon\, m^2}{4|\Theta \times \Theta' \times \mathcal{T}|^{1/2}},$$

which directly gives us the approximation result via the defition of N_{DOD}

$$\mathcal{E}_{\text{DOD}} < \frac{\varepsilon}{4} + m^{-1} \sup_{(\mu,t)\in(\Theta_{\text{data}}\times\mathcal{T}_{\text{data}})} \sqrt{\sum_{k > N_{\text{DOD}}} \sigma(\mu,t)_k^2} \le \frac{\varepsilon}{2}. \tag{4.9}$$

Include the cost via Yarotski

Argue the actual part for the NN error

Moreover we can use the exact same arguments as in the bound for $\mathcal{E}_{\text{NN}}$ for the POD-DL-ROM as in the proof of Theorem (4.8), which yields

$$\mathcal{E}_{\text{NN}} < \frac{\varepsilon}{4}. \tag{4.10}$$

Finally putting each Equation, i.e. (4.8), (4.9) and (4.10), together, we get

$$\mathcal{E}_R \leq \mathcal{E}_S + \mathcal{E}_{\text{DOD}} + \mathcal{E}_{\text{NN}} < \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon,$$

with probability greater than $1 - \delta$ concluding our proof. □

4.3 Comparison between existing Techniques

Chapter 5

Numerical Experiments

Include all numerical experiments

Find increase of performance depending on latent dimension

Give benchmark on pass forward times

Discuss comparison between different DL-ROMs

Chapter 6

Conclusion

Discuss results

Insert outlook for future ideas and implementation

Bibliography

Alt, H. W. (2016, July). *Linear functional analysis*. Springer London. https://doi.org/10.1007/978-1-4471-7280-2

Barrault, M., Maday, Y., Nguyen, N. C., & Patera, A. T. (2004). An 'empirical interpolation' method: Application to efficient reduced-basis discretization of partial differential equations. *Comptes Rendus Mathematique*, *339*(9), 667–672. https://doi.org/10.1016/j.crma.2004.08.006

Bauer, F. L., & Fike, C. T. (1960). Norms and exclusion theorems. *Numerische Mathematik*, *2*, 137–141. https://api.semanticscholar.org/CorpusID:121278235

Bramble, J. H., & Hilbert, S. R. (1970). Bounds for a class of linear functionals with applications to Hermite interpolation. *Numerische Mathematik*, *16*(4), 362–369. https://doi.org/10.1007/BF02165086

Brezis, H. (2011). *Functional analysis, sobolev spaces and partial differential equations* (1st ed.). Springer New York. https://doi.org/10.1007/978-0-387-70914-7

Brezis, H., & Browder, F. (1998). Partial differential equations in the 20th century. *Advances in Mathematics*, *135*(1), 76–144. https://doi.org/https://doi.org/10.1006/aima.1997.1713

Brivio, S., Fresca, S., Franco, N. R., & Manzoni, A. (2023). Error estimates for pod-dl-roms: A deep learning framework for reduced order modeling of nonlinear parametrized pdes enhanced by proper orthogonal decomposition. https://arxiv.org/abs/2305.04680

Chaturantabut, S., & Sorensen, D. C. (2010). Nonlinear model reduction via discrete empirical interpolation. *SIAM Journal on Scientific Computing*, *32*(5), 2737–2764. https://doi.org/10.1137/090766498

Ciampiconi, L., Elwood, A., Leonardi, M., Mohamed, A., & Rozza, A. (2024). A survey and taxonomy of loss functions in machine learning. https://arxiv.org/abs/2301.05579

Courant, R., & Hilbert, D. (1953). *Methods of mathematical physics, vol. i*. Interscience Publishers.

Dacorogna, B. (2004, November). *Introduction to the calculus of variations*. https://doi.org/10.1142/p361

Franco, N. R., Manzoni, A., Zunino, P., & Hesthaven, J. S. (2024). Deep orthogonal decomposition: A continuously adaptive data-driven approach to model order reduction [arXiv:2404.18841]. https://arxiv.org/abs/2404.18841

Fresca, S., Dede, L., & Manzoni, A. (2020). A comprehensive deep learning-based approach to reduced order modeling of nonlinear time-dependent parametrized pdes. https://arxiv.org/abs/2001.04001

Fresca, S., & Manzoni, A. (2022). Pod-dl-rom: Enhancing deep learning-based reduced order models for nonlinear parametrized pdes by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, *388*, 114181. https://doi.org/10.1016/j.cma.2021.114181

Golub, G. H., & Van Loan, C. F. (2013). *Matrix computations*. Johns Hopkins University Press. https://doi.org/10.56021/9781421407944

Grepl, M. A., & Patera, A. T. (2005). A posteriori error bounds for reduced-basis approximations of parametrized parabolic partial differential equations. *ESAIM: Mathematical Modelling and Numerical Analysis*, *39*, 157–181. https://doi.org/10.1051/m2an:2005006

Gühring, I., Kutyniok, G., & Petersen, P. (2019). Error bounds for approximations with deep relu neural networks in $W^{s,p}$ norms. https://arxiv.org/abs/1902.07896

Gühring, I., Raslan, M., & Kutyniok, G. (2020). Expressivity of deep neural networks. https://arxiv.org/abs/2007.04759

Haasdonk, B., & Ohlberger, M. (2008). Reduced basis method for finite volume approximations of parametrized linear evolution equations. *ESAIM: Mathematical Modelling and Numerical Analysis*, *42*, 277–302. https://doi.org/10.1051/m2an:2008001

Hesthaven, J., Pagliantini, C., & Rozza, G. (2022). Reduced basis methods for time-dependent problems. *Acta Numerica*, *31*, 265–345. https://doi.org/10.1017/S0962492922000058

Holmes, P., Lumley, J. L., & Berkooz, G. (1996). *Turbulence, coherent structures, dynamical systems and symmetry*. Cambridge University Press.

Hornik, K. (1991). Approximation capabilities of multilayer feedforward networks. *Neural Networks*, *4*(2), 251–257. https://doi.org/https://doi.org/10.1016/0893-6080(91)90009-T

Kunisch, K., & Volkwein, S. (2001). Galerkin proper orthogonal decomposition methods for a general equation in fluid dynamics. *SIAM Journal on Numerical Analysis*, *40*(2), 492–515. https://doi.org/10.1137/S0036142900382612

Lumley, J. L. (1967). The structure of inhomogeneous turbulent flows. In A. M. Yaglom & V. I. Tatarsky (Eds.), *Atmospheric turbulence and radio wave propagation* (pp. 166–178). Nauka.

Maday, Y., Patera, A. T., & Rovas, D. V. (2002). A blackbox reduced-basis output bound method for noncoercive linear problems. In *Proceedings of the 12th international conference on domain decomposition methods*. Domain Decomposition Press.

Noor, A. K. (1980). Recent advances in reduction methods for nonlinear problems. *Computers & Structures*, *12*(1), 31–44. https://doi.org/10.1016/0045-7949(80)90101-6

Patera, A. T. (1984). A spectral element method for fluid dynamics: Laminar flow in a channel expansion. *Journal of Computational Physics*, *54*(3), 468–488. https://doi.org/10.1016/0021-9991(84)90128-1

Peherstorfer, B., Willcox, K., & Gunzburger, M. (2016). Optimal model management for multifidelity monte carlo estimation. *SIAM Journal on Scientific Computing*, *38*(5), A3163–A3194. https://doi.org/10.1137/15M1046472

Prud'homme, C., Rovas, D. V., Veroy, K., Machiels, L., Maday, Y., Patera, A. T., & Turinici, G. (2002). Reliable real-time solution of parametrized partial differential equations: Reduced-basis output bound methods. *Journal of Fluids Engineering*, *124*(1), 70–80. https://doi.org/10.1115/1.1448332

Quarteroni, A., Manzoni, A., & Negri, F. (2015). *Reduced basis methods for partial differential equations*. Springer. https://doi.org/10.1007/978-3-319-15431-2

Rowley, C. W., Mezić, I., Bagheri, S., Schlatter, P., & Henningson, D. S. (2009). Spectral analysis of nonlinear flows. *Journal of Fluid Mechanics*, *641*, 115–127. https://doi.org/10.1017/S0022112009992059

Schmid, P. J. (2010). Dynamic mode decomposition of numerical and experimental data. *Proceedings of the 61st Annual Meeting of the APS Division of Fluid Dynamics*.

Siena, P., Girfoglio, M., & Rozza, G. (2022). An introduction to pod-greedy-galerkin reduced basis method. https://arxiv.org/abs/2203.08532

Sirovich, L. (1987). Turbulence and the dynamics of coherent structures. part i: Coherent structures. *Journal of Fluid Mechanics*, *175*, 419–427. https://doi.org/10.1017/S0022112087000199

Yarotsky, D. (2017). Error bounds for approximations with deep relu networks. *Neural Networks*, *94*, 103–114. https://doi.org/https://doi.org/10.1016/j.neunet.2017.07.002